

Chapter 95 — Urinary Lithiasis: Etiology, Epidemiology, and Pathogenesis

- **Stone disease is ancient but has changed its site and its sex:** with Westernisation, stone formation has migrated from the lower to the upper urinary tract, and a disease that historically favoured males is becoming increasingly sex-neutral.
- **Minimally invasive and non-invasive techniques** have transformed stone *removal*, but **surgery does not alter the course of the disease** — the stone is removed, the tendency to form stones is not.
- **Cost:** annual United States (US) expenditure on insured patients with nephrolithiasis **exceeded \$10 billion in 2012**, a nearly fivefold rise since 2000.
- Because **recurrence is frequent**, a **medical prophylactic programme** is desirable, and that requires understanding of cause, epidemiology and pathogenesis.

Concept — *Stone treatment is two separate problems: removal is surgical, recurrence is metabolic, and the rest of this chapter exists to explain the metabolic half.*

EPIDEMIOLOGY OF RENAL CALCULI

Prevalence and Incidence

- **Lifetime prevalence 1–15%**, varying with age, sex, race and geography.
- **Regional prevalence:** North America 7–13%, Europe 5–9%, Asia 1–5%.
- **National Health and Nutrition Examination Survey (NHANES):** linear rise in US adult prevalence 1976–2010; a transient dip (2007–2008 to 2011–2012) was followed by renewed rise to **about 10% in 2015–2016**; age-standardised prevalence 2007–2016 $\approx$ 9.3%.
- **12-month incidence (2015–2018) $\approx$ 2.1%**, about 2000 stones per 100,000 adults.
- **The rise is global** (Europe, Japan, US) and **partly a detection artefact** — more **computed tomography (CT)** finds more **asymptomatic calculi**.

Sex

- **Historically male-predominant:** by inpatient, outpatient and emergency-department indicators, males were affected **two to three times** more often than females.
- **The gap is narrowing:**
 - US inpatient discharges 1997–2002: **female discharges rose, male discharges fell**; the **male-to-female ratio fell from 1.7 to 1.3**.
 - NHANES 2007–2018: **prevalence rose in females, unchanged in males** — persisting after adjustment for sociodemographic and dietary factors.
 - Population data 1997–2012: **female incidence rose about 15% per 5 years; male incidence stable**; the **steepest female rise was in 10- to 19-year-olds**.
- **Females carry more stone-related morbidity** (retrospective data): more **positive urine cultures**, **nephrostomy placement**, **intensive care admission** and **longer stays**.

Race and Ethnicity

- **NHANES 2007–2016 prevalence by group** (highest to lowest): **non-Hispanic White > Hispanic > non-Hispanic Black > non-Hispanic Asian**.
- **Trend:** prevalence **rose in White and (slightly) Hispanic groups**; **stable in Black and Asian groups**.

- **Sex ratio varies by race** (table); rising incidence was **greater in Black females than Black males**.
- **Race is a social construct, not a genetic proxy**: differences reflect **genetics plus cultural and sociodemographic factors**.

Governing contrast	White	Black
Prevalence rank	Highest	Low (second lowest)
Male-to-female ratio	≈ 2.3 (male predominance)	≈ 0.65 (female predominance)

Age

- **Paediatric stones are rising at 5–10% per year**; historically uncommon under age 20.
- **Paediatric incidence peaks in female adolescents** — the **reverse of the adult sex pattern**; young females are also more often **hospitalised**.
- Population data (Olmsted County, 1984–2008): overall paediatric rise **about 4% per year, about 6% per year in 12- to 17-year-olds**.
- **Proposed drivers of the paediatric rise: obesity and metabolic syndrome** reaching younger generations, or changes in the **exposome** rather than genetics or comorbidity alone.
- **Adult incidence peaks in the fourth to sixth decades**; prevalence rises with age in both sexes — in males highest over 80 years.
- **Females have lower prevalence than males at every age except 20–39 years**, where female prevalence is higher.
- Recent NHANES: female prevalence **rose only in those under 60; no rise in females over 60**.
- **Bimodal female distribution**: a **second peak in the sixth decade at menopause**, attributed to **loss of oestrogen**.
 - **Oestrogen is protective** — it **enhances renal calcium reabsorption** and **reduces bone resorption**.
 - Females have **lower urinary calcium and calcium oxalate and brushite saturation** than males **until beyond age 50**; **oestrogen-treated postmenopausal females** have lower urinary calcium and saturation than untreated ones.

Concept — *Age and sex patterns are largely hormonal and metabolic: oestrogen holds urinary calcium down until menopause, while obesity and metabolic syndrome are pulling stone disease into younger, and increasingly female, populations.*

Geography and Climate

- **Prevalence follows environmental risk**: highest in **hot, arid or dry climates** (mountains, desert, tropics), although **genetic and dietary factors may outweigh geography**.
- **US veteran data**: calcium oxalate stones **highest in the Southeast**, uric acid stones **highest in the East**.
- **Prevalence rises from north to south and from west to east, highest in the Southeast** (the "stone belt"); **ambient temperature and sunlight are independent predictors**.
- **Seasonal variation**: **peak in summer (July–September)**, within **1–2 months of maximal mean temperature** — via **perspiration fluid losses** and possibly **sunlight-induced vitamin D**. Confirmed worldwide by **systematic review**.
- **Short latency**: once mean daily temperature exceeds **10 °C**, stone presentations rise **within the ensuing 10 days**.
- **Military translocation studies** (a natural experiment):
 - Soldiers in desert postings — **more renal colic in summer**.
 - Personnel deployed to Kuwait and Iraq — **mean time to stone formation ≈ 93 days**.

- Soldiers transferred to the Persian Gulf **in summer** developed **raised urinary calcium within 10 days** (not in the cold season) — attributed to **sunlight-induced 1,25-dihydroxyvitamin D3 [1,25(OH)2D3]**.
- **Heat response is mainly male:** prevalence rises more steeply per degree in males, and heat-related emergency presentation rises **more in males** and most at **40–69 years** (confounders: sunlight, occupation, hydration).
- **Global warming projections:** modelled temperature–prevalence relationships (**peaked/non-linear** in one data set, **linear** in another) predict **1–1.5 million additional lifetime cases by 2050** and a **northward expansion of the stone belt**; a further projection anticipates **about 10,000 additional emergency presentations between 2025 and 2089**.
- **Urban heat islands:** urbanisation, architecture and reduced vegetation make cities warmer — a proposed contributor, **not yet tested** against fluid intake, diet and socioeconomic status.

Concept — *Climate acts indirectly — heat drives low urine volume and sunlight drives vitamin D-mediated hypercalciuria — so geography is really a proxy for these two urinary effects, predominantly in males.*

Occupation

- **Heat exposure and dehydration** are occupational risk factors:
 - Royal Navy — **cooks and engine-room personnel** had the **highest stone rates**.
 - **Steelworkers in hot areas** — markedly higher stone incidence than those at normal temperatures, with **low urine volume and hypocitraturia**.
 - **Glass-plant workers exposed to heat** — **lower urine volume and pH, higher uric acid and specific gravity**, hence **high uric acid saturation**; stone formers among them had a **strikingly high proportion of uric acid stones**.
- **Sedentary occupations (managerial, professional)** carry **increased risk**, reason unclear:
 - consistent with higher risk in **affluent individuals and societies** (richer diet and lifestyle);
 - alternatively, **inactivity increases bone turnover and urinary calcium** — the same mechanism seen in **astronauts under weightlessness**, where **exercise** to limit bone demineralisation is the best prevention.
- **Restricted bathroom access (taxi drivers, operating-room staff)** increases risk.

Occupational mechanism	Examples	Urinary consequence
Heat / dehydration	Cooks, engine rooms, steel and glass workers	Low volume, low pH, hypocitraturia → uric acid stones
Sedentary	Managerial, professional; astronauts (weightlessness)	Bone turnover → hypercalciuria
Restricted voiding / fluid access	Taxi drivers, operating-room staff	Low volume

Obesity, Diabetes, and Metabolic Syndrome

- **Stone prevalence and incident risk rise with weight and body mass index (BMI)** in both sexes — **the association is stronger in females** (two large prospective cohorts).
- **Obesity and weight gain are independent risk factors, not explained by diet alone**; confirmed in further cohorts and claims data.
- **Women's Health Initiative Observational Study** (postmenopausal females without prior stones): **higher BMI, less physical activity and energy intake > 2200 kcal/day** each independently raised incident risk — supporting **diet and exercise components of weight-loss counselling** in prevention.

- **Metabolic syndrome** (visceral obesity with hyperlipidaemia, hypertriglyceridaemia, hyperglycaemia and/or hypertension) increases stone risk, with **risk rising with the number of metabolic traits**; it is also a precursor of **type 2 diabetes mellitus**.
- **Diabetes and stones — a bidirectional, sex-specific link:**
 - **Diabetes predicts incident stones in females but not males** (after adjusting for BMI, diet and thiazides).
 - **A stone history predicts incident diabetes in both sexes**; stone formers are about **1.3 times more likely** to be diagnosed with diabetes within 5 years.
 - In asymptomatic stones followed by CT, **fasting glucose ≥ 100 mg/dL predicted stone growth** — suggesting **glycaemic control may reduce recurrence**.
- **Unifying mechanism: insulin resistance:**
 - **Obesity and insulin resistance $\rightarrow$ low urine pH $\rightarrow$ uric acid stones.**
 - **Hyperinsulinaemia $\rightarrow$ hypercalciuria $\rightarrow$ calcium stones.**
 - **Overweight (BMI 25–30 kg/m²)** is already associated with **lower urine pH** and **higher urinary sodium, uric acid and calcium**.
 - **Abnormalities are harder to correct with rising BMI:** response to **potassium citrate** for low pH and hypocitraturia falls as BMI rises, requiring **more dose adjustment**.
- **Hepatic link to oxalate:** in a mouse model, **non-alcoholic steatohepatitis (NASH)** down-regulated **alanine-glyoxylate aminotransferase (AGXT)** — the enzyme deficient in **primary hyperoxaluria type 1 (PH1)** — increasing oxalate production; a proposed explanation for **calcium oxalate stones in diabetes, obesity and metabolic syndrome**.
- **Higher BMI $\rightarrow$ more urinary oxalate, uric acid, sodium and phosphorus.**

Obesity-associated stone	Principal driver
Calcium oxalate	Increased excretion of promoters (oxalate, calcium, sodium, uric acid)
Uric acid	Low urinary pH

Concept — *Obesity, diabetes and metabolic syndrome converge on insulin resistance, which lowers urine pH (uric acid stones) and raises promoter excretion (calcium oxalate stones) — so the stone is a urinary marker of a systemic metabolic state.*

Cardiovascular Disease

- **Hypertension: meta-analysis** shows stone formers have a **significantly increased risk of hypertension, stronger in females**.
 - Proposed shared dietary exposures: **calcium, sodium, potassium**.
 - Hypertensive individuals have **higher urinary calcium, uric acid, oxalate and calcium oxalate supersaturation**; hypertensive stone formers excrete **about 25 mg/day more calcium** than normotensive stone formers.
- **Heart disease — association, causation unproven:**
 - Stone history associated with **higher incidence of myocardial infarction (MI)**, independent of **chronic kidney disease (CKD)**.
 - Stone history associated with **subclinical carotid atherosclerosis** in young people.
 - In three large prospective cohorts, stones predicted **heart disease in both female cohorts but not the male cohort**.

- Large population cohort (> 3 million): stone formers had **higher risk of acute MI and stroke, greater in younger individuals (< 50 years) and in females.**
- **Candidate mechanisms:**
 - **Dyslipidaemia** (high total cholesterol, high triglycerides, low **high-density lipoprotein (HDL)**) alters urine chemistry towards stone formation.
 - **Calcium-sensing receptor (CaSR)** dysfunction — a G-protein-coupled receptor expressed in **kidney and vasculature** — a molecular bridge.
 - **Oxidative stress, endothelial dysfunction and inflammation.**

Chronic Kidney Disease

- **Stone history is associated with CKD** after controlling for confounders (case-control study).
- Stone formers, **especially with BMI ≥ 27 kg/m²**, have a **lower estimated glomerular filtration rate (eGFR)** and are **nearly twice as likely to have stage III CKD.**
- **Prospective cohorts** (insured, predominantly White) confirm **higher incident CKD.**
- **Stones associated with CKD and end-stage renal disease (ESRD) requiring dialysis** — accounted for largely by females; no association in males.
- **Symptomatic stone formers have increased ESRD risk**, possibly from **repeated obstruction.**
- **Causal versus shared-risk-factor relationship remains unresolved.**

Concept — Across hypertension, cardiovascular disease and CKD the pattern repeats: the association is real, stronger in females, and not proven causal — stones and these diseases probably share a metabolic soil.

Water

- **Fluid intake is inversely related to incident stones** (two large observational studies).
- **Landmark trial — Borghi (1996)**, randomised trial in **first-time idiopathic calcium stone formers**: high fluid intake raised urine volume and cut **recurrence to 12% versus 27%** in controls given no advice.
- **Water hardness** (calcium carbonate content):
 - Some studies found **fewer stones in "hard"-water regions**; others found **no difference.**
 - Hardness correlates with **urinary calcium and citrate**, but **not with stone episodes.**
 - **Systematic review (1989–2018)**: hard water and mineral water raise calcium intake (and potentially urinary calcium), yet the **net effect is reduced calcium stone formation**, putatively from **high magnesium and bicarbonate** — but studies were **heterogeneous, no meta-analysis**, so **interpret with caution.**

Hard water — opposing effects	Direction
High calcium content	Raises urinary calcium (pro-stone)
High magnesium and bicarbonate	Inhibit / raise citrate (anti-stone) — net effect probably protective

Concept — Volume is the one universally effective modifier — proven by randomised trial — whereas water composition is a weak, conflicting and poorly evidenced influence.

KEY POINTS: EPIDEMIOLOGY

- **Upper-tract stones are commoner in males, but the gap is narrowing.**
- **White individuals have the highest incidence** compared with Asian, Hispanic and Black individuals.
- **Geographic variability**, with **highest prevalence in the Southeast (US).**
- **Risk correlates with weight and BMI.**

- Stones are associated with **diabetes, metabolic syndrome, cardiovascular disease and CKD, causation unproven.**

PHYSICOCHEMISTRY AND PATHOGENESIS

- **Stone formation is a cascade along the nephron:** urine becomes **supersaturated** → ions **precipitate as crystals or nuclei** → crystals either **pass in the urine** or are **retained at anchoring sites** → **growth and aggregation** → stone.

State of Saturation

- **Concentration product (CP)** — the product of the concentrations of a salt's pure chemical components (e.g. $CP = [Na^+][Cl^-]$).
- **Thermodynamic solubility product (Ksp)** — the CP at **saturation**, where dissolved and crystalline phases are in **equilibrium**; adding crystals beyond this causes precipitation unless **pH or temperature** change.
- **Formation product (Kf)** — the CP at which the salt **can no longer be held in solution** and crystals form spontaneously.
- **Urine holds salts above Ksp without precipitating** because of **inhibitors and other molecules** — this supersaturated-but-stable state is **metastable**.

State (Fig. 95.3)	Concentration product	What happens
Undersaturated	< Ksp	Crystals never form; existing stones may dissolve
Metastable	Ksp to Kf — where most urinary stone salts sit	No spontaneous (homogeneous) nucleation; growth on existing crystals possible; heterogeneous nucleation may occur; inhibitors effective; matrix may be involved; the zone of therapeutic intervention
Unstable	> Kf	Spontaneous nucleation, growth and aggregation; inhibitors not generally effective

- **How crystals still form in metastable urine** — three routes:
 1. **Local CP exceeds Kf** in parts of the nephron long enough for nucleation.
 2. **Obstruction or stasis** prolongs transit time.
 3. **Heterogeneous nucleation** — impurities adsorb crystal components in a lattice-like geometry, needing much less energy than homogeneous nucleation.
- **Estimating saturation:**
 - **Activity product ratio** — accounts for **urine pH and ionic activities** of all major ion species and overall ionic strength.
 - **EQUIL 2** — computer program, **commonly used**; gives the **relative saturation ratio (RSR)**, also called **concentration product ratio (CPR)** = urine CP ÷ Ksp of the salt.
 - **RSR < 1** → crystals dissolve; **RSR > 1** → crystals form and grow.
- **Lowering RSR:**
 - **Reduce urinary concentration** of stone components (calcium, oxalate), **reduce filtered load**, or **increase tubular reabsorption**.
 - **Complexation** — **citrate binds calcium**, cutting free ionic calcium.
 - **pH manipulation works for phosphate** (strongly pH-dependent) but **not for oxalate** — **oxalic acid is a strong acid (pK ≈ 4)**, so physiological pH changes barely alter oxalate.

Saturation method	Basis	Performance against direct measurement
EQUIL 2 → RSR	Computer-derived ionic activities	Overestimates saturation ($\approx 80\%$ for brushite, $\approx 50\%$ for calcium oxalate)
JESS (Joint Expert Speciation System) → supersaturation index (SI)	Adds soluble complexes EQUIL 2 omits (dicalcium-dihydrogen phosphate, calcium phosphocitrate — pH- and citrate-dependent); therefore lower estimated ionised calcium, phosphate and oxalate	No significant difference from direct measurement — the more reliable routine estimate
Experimentally determined CPR	CP measured before and after incubation with synthetic stone salt (growth or dissolution) — no computed activities	Reference method , but too labour-intensive for routine use

• **Calcium versus oxalate — which matters more?**

- **Historical view: oxalate dominates** — a rise in oxalate raised calcium oxalate saturation more than a rise in calcium, and **calcium-driven saturation plateaued below K_f** whereas oxalate-driven saturation did not.
- **Revised view: equal contributors** — the relative effect depends on the **stability constant** chosen; with the accepted constant (2.746×10^3 , used in EQUIL 2) **calcium and oxalate have equivalent effects** (Fig. 95.4). Re-analysis with JESS confirmed equivalence, though dependence was less marked.
- **Therapeutic corollary: lowering either calcium or oxalate** reduces calcium oxalate saturation.

Concept — Every prevention strategy is an attempt to push urine down out of the metastable zone — by dilution, by lowering calcium or oxalate (equally effective), by complexation with citrate, or (for phosphate and uric acid, never oxalate) by changing pH.

Nucleation and Crystal Growth, Aggregation, and Retention

- Normal urine holds calcium oxalate at about four times its aqueous solubility.
- Promoters of calcium oxalate supersaturation: low volume, low citrate; high calcium, oxalate, phosphate, uric acid.
- With inhibitors present, calcium oxalate precipitates only at 7–11 times its solubility.
- Homogeneous nucleation — nuclei form in pure solution; **nuclei = earliest crystal structures that will not dissolve**.
 - Small nuclei are unstable — below a **critical size**, dissolution is favoured.
 - Nuclei **persist** if driving force (supersaturation) and stability are favourable and **lag time to nucleation is short relative to nephron transit**.
 - Inhibitors (e.g. citrate) **destabilise nuclei**; promoters **stabilise them** by offering a binding surface.
- In vivo, nucleation is usually heterogeneous — on epithelial cells, cell debris or other crystals.
- Nephron transit $\approx 5\text{--}7$ minutes — too short for a single crystal to grow large enough to occlude a tubule, but **aggregation** of many nuclei can produce occluding particles **within minutes**.
- Inhibitors can **block growth or aggregation** at each step (see the inhibitor table).

Theory of retention	Free particle	Fixed particle
Premise	Crystals grow and aggregate free in tubular fluid , then are retained	Crystals bind an anchoring site , prolonging exposure to supersaturated urine

Theory of retention	Free particle	Fixed particle
Status	Initially judged impossible within transit time ; recalculation with current nephron dimensions shows retention is possible	Anchoring site = Randall plaque on the renal papilla
Histological support	Crystals occluding inner medullary collecting ducts (ductal plugs) in selected patients	Subepithelial papillary plaque in idiopathic calcium oxalate stone formers

Randall Plaque

- **Randall (1937)** found **subepithelial papillary plaque in 17% of autopsy kidneys**; proposed an origin in the **basement membrane**, with **erosion** through the papillary surface creating a **nidus**.
- **Endoscopy**: papillary plaques in **74% of stone formers versus 43% of controls**.
- **Papillary biopsy at percutaneous nephrolithotomy** in idiopathic calcium oxalate stone formers localised the sequence: **basement membrane of the thin limbs of the loop of Henle → medullary interstitium → subepithelial papilla → erosion through urothelium → anchored surface → calcium oxalate stone grows attached**.
- **Plaque mineral = calcium apatite (carbapatite)** — by infrared microspectroscopy, electron diffraction and analysis of > 5000 plaque-associated stones.
- **Microstructure: laminated particles** of alternating mineral and organic layers; **osteopontin** sits at the crystal-organic interface; apatite forms **spherulites** that grow, aggregate and fuse; in the basement membrane they lie in amorphous matrix and **membrane-bound vesicles**; migrating through the interstitium they propagate a **mineralisation front** on **collagen and cell debris** until they reach the papillary subepithelium.
- **Calcifying nanoparticles ("nanobacteria")** — implicated in atherosclerotic calcification; **associated (not proven causal)** with Randall plaque.
- **Vascular hypothesis**: injury to the **vasa recta** near the papilla → **atherosclerosis-like repair and calcification** → erosion into interstitium and collecting ducts → nidus.
 - Supported by **osteogenic transformation**: vascular smooth muscle becomes osteoblast-like with high calcium and phosphate; **rat tubular cells show osteogenic activity** with high oxalate or crystals, mediated by **nicotinamide adenine dinucleotide phosphate (NADPH) oxidase** and **reactive oxygen species**.

Plaque versus Plug — by Stone-Former Type

Stone former	Randall plaque	Ductal plugs	Other histology
Idiopathic calcium oxalate	Present — the anchoring site	Uncommon (but see below)	—
Intestinal bypass (enteric hyperoxaluria)	Absent	Apatite plugs in inner medullary collecting ducts	Epithelial damage, interstitial inflammation and fibrosis
Brushite	Present (interstitial apatite)	Apatite plugs in inner medullary and terminal collecting ducts	Collecting-duct injury, interstitial fibrosis — intermediate between the two above

Stone former	Randall plaque	Ductal plugs	Other histology
Distal renal tubular acidosis (RTA), primary and secondary hyperparathyroidism, cystinuria	—	Ductal plugs present	—

- **Brushite pathogenesis (proposed):** apatite crystallises in collecting ducts → duct injury, cell death, duct dilatation → interstitial inflammation → progressive involvement of adjacent tissue.
- **Calcium oxalate-to-brushite conversion hypothesis:** an insult such as infection or shock-wave lithotripsy causes tubular dysfunction, alkaline environment, inflammation and intraductal hyaluronic acid, converting a calcium oxalate former into a brushite former.
- **Evidence that idiopathic stones start on plaque:** plaque area correlates with stone number and 24-hour urine parameters; papillary mapping and micro-CT show stones originating on interstitial plaque.
- **The distinction has blurred** (consecutive endoscopic series):
 - Randall plaque in almost all stone formers (99%) — including uric acid formers.
 - Ductal plugs in half of uric acid formers and in up to a third of idiopathic calcium oxalate formers (severe in about a tenth), correlating with urine risk parameters.
 - Plaque covered less papillary surface than earlier reports and did not correlate with urine parameters — possibly because this cohort was more metabolically diverse (not restricted to hypercalciuria).
 - Practical rule: ductal plugs should raise suspicion of a secondary cause.
- **"Unified" theory of plaques and plugs:**
 - **Plaque:** oxidative stress from excess urinary calcium, oxalate and phosphate and/or deficient inhibitors → tubular cells de-differentiate into osteoblast-like cells → calcium phosphate crystals in membrane-bound vesicles → breach basement membrane → contact collagen and membrane debris → plaque spreads through interstitium with inflammation and fibrosis → matrix metalloproteinases breach papillary epithelium → plaque exposed to pelvic urine → stone growth.
 - **Plugs:** supersaturated tubular fluid and pelvic urine form plugs that slow outflow from ducts, promoting crystal formation and aggregation by retention.
- **Knowledge will remain incomplete without a reliable animal model.**

Concept — Two routes to a stone: plaque (an interstitial apatite anchor on which calcium oxalate grows — the idiopathic pathway) and plug (intratubular crystal obstruction with tubular injury — the pathway of secondary, systemic disease); finding plugs should prompt a search for the secondary cause.

Inhibitors and Promoters of Crystal Formation

- **Urine is normally supersaturated** for calcium, oxalate and phosphate; stones do not form routinely because inhibitors raise the supersaturation needed for nucleation or slow growth and aggregation.
- **Inhibitors exist for calcium oxalate and calcium phosphate** — none is known for uric acid.
- **Blocking crystal adhesion to epithelium** can also prevent retention and growth.
- **Whole urine raises the formation product of calcium phosphate;** inorganic pyrophosphate accounts for a substantial share (25–50%) of this inhibitory activity by one method; by another, citrate, magnesium and pyrophosphate together account for a smaller share, citrate being the most important.

Inhibitor	Nature / source	Principal action	Distinguishing feature
Citrate	Small organic anion	(1) Complexes calcium ; (2) inhibits spontaneous calcium oxalate precipitation and agglomeration ; (3) prevents heterogeneous nucleation of calcium oxalate by monosodium urate	Weak against calcium oxalate growth , potent against calcium phosphate growth
Magnesium	Divalent cation	Complexes oxalate → lowers ionic oxalate and calcium oxalate supersaturation	Synergy with citrate (negated by uric acid) and with phytate
Phytate	Dietary	Inhibitor in synthetic urine	Synergistic with magnesium
Glycosaminoglycans (and acid mucopolysaccharides, RNA)	Polyanions	Bind surface calcium → inhibit nucleation and growth	Chondroitin sulphate most abundant ; heparan sulphate binds calcium oxalate monohydrate most strongly ; lower in stone formers , lowest in recurrent formers ; yet incorporated into apatite stone lattice
Nephrocalcin	Acidic glycoprotein; proximal tubule and thick ascending limb	Inhibits calcium oxalate monohydrate growth , nucleation and aggregation	Four isoforms : non-formers excrete the two γ -carboxylglutamic acid-containing, inhibitory isoforms; stone formers excrete the two non-inhibitory isoforms
Tamm-Horsfall protein (uromodulin)	Membrane-anchored; thick ascending limb and distal convoluted tubule ; released by phospholipase/ protease cleavage	Inhibits calcium oxalate monohydrate aggregation , not growth	Most abundant urinary protein ; inhibitor in alkaline urine , promoter (polymerises) in acidic urine ; a uromodulin allele protects against calcium stones (genome-wide association study, GWAS); may regulate TRPV5 and TRPV6 calcium channels
Osteopontin (uropontin)	Acidic phosphorylated glycoprotein; bone matrix and ascending limb/distal tubule	Inhibits nucleation, growth, aggregation and crystal binding to epithelium	Inducible inhibitor working with constitutive Tamm-Horsfall protein
Urinary prothrombin fragment 1 (F1)	Crystal matrix protein resembling prothrombin F1	Reduces aggregation and deposition	—
Matrix Gla protein (MGP)	Vitamin K-dependent ; inhibits vascular calcification	—	Gene polymorphism linked to vascular calcification and MI; a single nucleotide polymorphism (SNP) linked to calcium oxalate stones in Japanese and Chinese patients

Inhibitor	Nature / source	Principal action	Distinguishing feature
Bikunin	Light chain of inter-α-trypsin inhibitor (hepatic glycoprotein: two heavy, one light chain)	Strong inhibitor of calcium oxalate crystallisation, growth and aggregation in vitro	Up-regulated by oxalate exposure (rat)

Concept — Stone formation is a balance between supersaturation and inhibition; most inhibitors act on calcium salts by complexation (citrate–calcium, magnesium–oxalate) or surface binding (glycoproteins), and the absence of any uric acid inhibitor is why uric acid stones depend almost entirely on pH.

Matrix

- **Matrix = the non-crystalline component** of stones — $\approx$ 2.5% of stone weight.
- **Matrix-dominant stones** (a majority of the stone, up to about 65%) occur with **chronic urinary tract infection**.
- **Composition** hard to define (**only about a quarter is soluble**): roughly **two-thirds protein**, with **non-amino sugars, glucosamine, bound water and organic ash**.
- **Proteins in matrix**: Tamm-Horsfall protein, nephrocalcin, a γ -carboxyglutamic acid-rich protein, renal lithostathine, albumin, glycosaminoglycans, free carbohydrates and matrix substance A.
- **Matrix substance A** — **immunologically unique mucoprotein** present in the matrix of **all** stone formers; its antigens are **found in the urine of most stone formers and of no healthy individuals**.
- **Proteomics**:
 - Dozens of proteins in calcium oxalate stones, **many inflammatory; inflammatory proteins predominate in both calcium oxalate and calcium phosphate stones** — suggesting a **shared inflammatory pathogenesis**.
 - Later analysis added **osteopontin, prothrombin F1 and Tamm-Horsfall protein plus leucocyte proteins (myeloperoxidase, lactotransferrin)**; variability between stones suggests **diverse, partly hereditary aetiology**.
- **Role of matrix** — **promoter, inhibitor or bystander** — is **undetermined**.

Figure legends

- **Fig. 95.7 — Brushite stone formers**: (A) white subepithelial plaque plus a **yellowish crystalline deposit at the openings of the ducts of Bellini**; (B) biopsy — **large amounts of Yasue-positive material in the ducts of Bellini**. (From Evan et al., 2005.)

KEY POINTS: PHYSICOCHEMISTRY AND PATHOGENESIS

- **Urine must be supersaturated for stones to form.**
- **Supersaturation alone is not sufficient** — urinary inhibitors prevent crystallisation.
- **Nephrocalcin, uropontin and Tamm-Horsfall protein** are important inhibitors of nucleation, growth and aggregation.
- **Urinary calcium and oxalate contribute equally** to calcium oxalate saturation.
- Common calcium stones may originate on **subepithelial calcium apatite plaques** that anchor calcium oxalate growth.
- **Matrix** — the non-crystalline component — comprises **mucoproteins, proteins, carbohydrates and urinary inhibitors**.

MINERAL METABOLISM

Calcium

Intestinal absorption

- 30–40% of dietary calcium is absorbed, mostly in the **small intestine**; only about 10% in the colon.
- Intestinal adaptation: fractional absorption rises when intake is low and falls when intake is high.

Pathway	Transcellular	Paracellular
Saturable?	Saturable	Non-saturable
Regulation	Vitamin D (calcitriol)-dependent — calcitriol increases brush-border calcium permeability	Unregulated, concentration-driven
Dominates when	Calcium intake is low; down-regulated by a calcium-replete diet	Calcium intake is high

- **Divided doses are absorbed better than one large dose** (because of the saturable component).
- **Net absorption:** a small amount of calcium is **secreted into the lumen**, so of a typical 600–1200 mg/day intake, 100–300 mg is absorbed net.
- Calcium is absorbed only in ionic form; luminal complexing agents reduce absorption — phosphate, citrate, oxalate, sulphate, fatty acids.
 - Calcium phosphate complexation is pH-dependent ($pK \approx 6.1$) — high luminal pH favours it.
 - Calcium oxalate complexation is less pH-dependent and less reversible — hence an oxalate-rich diet reduces calcium absorption.

The vitamin D–parathyroid axis

- 1,25(OH)₂D₃ (calcitriol) is the most potent stimulator of intestinal calcium absorption.
- Synthesis sequence: 7-dehydrocholesterol (skin) → sunlight → previtamin D₃ → liver 25-hydroxylation → 25-hydroxyvitamin D₃ → proximal renal tubule 1 α -hydroxylation → 1,25(OH)₂D₃.
- 1 α -hydroxylase (mitochondrial, proximal tubule) is **stimulated by parathyroid hormone (PTH)** and by hypophosphataemia.
- **Loop:** low serum calcium → PTH ↑ → 1 α -hydroxylase ↑ → calcitriol → vitamin D receptor (VDR) in brush border → gut calcium absorption ↑; calcitriol with PTH also recruits osteoclasts; rising serum calcium then suppresses PTH and calcitriol, and calcitriol inhibits PTH synthesis by up-regulating parathyroid VDR and CaSR.
- **PTH:**
 - 84-amino-acid peptide cleaved from **prepro-PTH**; only mature PTH is secreted.
 - **Most potent secretory stimulus:** fall in serum calcium, sensed by the **G-protein-coupled CaSR**, which also regulates renal tubular calcium reabsorption.
 - Acts via **cyclic adenosine monophosphate (cAMP)** and **phospholipase C**.
 - **Bone:** osteoclastic calcium and phosphorus release. **Kidney:** ↑ calcium reabsorption, ↓ phosphate reabsorption, ↑ calcitriol. **Gut:** no direct effect — only via calcitriol.

Renal calcium handling

Segment	Share of filtered calcium reabsorbed	Route	Key molecules
Proximal tubule	60–65% (the largest share)	Paracellular	Tight-junction proteins occludin , junctional adhesion molecules , claudins ; claudin-2 mediates cation permeation
Thick ascending limb (TAL)	25–30%	Paracellular, driven by a lumen-positive transepithelial voltage (from apical potassium secretion, basolateral chloride exit and the transepithelial sodium chloride gradient)	Claudin-16/claudin-19 complex forms the cation channel; CaSR → claudin-14 , which blocks the 16/19 channel
Distal convoluted tubule (DCT)	8–10% (fine regulation)	Transcellular	Transient receptor potential vanilloid 5 (TRPV5) apical entry (highly calcium-selective) → calbindin-D28k chaperone → basolateral exit

- Only 1–3% of filtered calcium is excreted.
- TRPV5 is regulated by gene expression, feedback inhibition and membrane trafficking; TRPV5 knockout mice develop severe hypercalciuria, compensated by calcitriol-driven intestinal hyperabsorption.

Concept — Calcium balance is a three-organ loop (gut, bone, kidney) closed by PTH and calcitriol: bulk renal reabsorption is paracellular and passive (proximal tubule, TAL), fine control is transcellular (TRPV5 in the DCT) — and a defect at any organ can produce hypercalciuria, which is why hypercalciuria was later classified by organ.

Phosphorus

- Absorption is saturable (transcellular) at low phosphate concentrations (1–3 mmol/L) and non-saturable (paracellular) at higher levels.
- About 60% of dietary phosphate is absorbed; active uptake is sodium-dependent and calcitriol-regulated.
- Strongly pH-dependent: low luminal pH reduces, high pH enhances phosphate absorption.
- About 65% of absorbed phosphate is excreted renally, the rest by the gut.
- Kidney reabsorbs 80–90% of filtered phosphate, excreting 10–20%.
- Primary regulator: PTH — inhibits tubular phosphate reabsorption.

Magnesium

- Absorbed by passive diffusion (predominant) and active transport, in large and small intestine — mainly the distal small intestine.
- Hormonal regulation mainly via vitamin D.

Oxalate

Contrast	Calcium	Oxalate
Fraction of intake absorbed	30–40%	6–14% (range between individuals 10–72%)
Main site	Small intestine (colon ~10%)	Throughout the gut — roughly half small intestine, half colon

Contrast	Calcium	Oxalate
Intake-absorption relationship	Adaptive (fractional absorption falls with high intake)	Curvilinear — higher fractional absorption at low intake

- **Urinary oxalate as a surrogate for absorption** is valid only if ingestion-excretion is linear and absorbed oxalate is not taken up, metabolised or re-secreted — assumptions that **do not strictly hold**.
- **Whether stone formers absorb more oxalate than non-formers is unclear.**
- **Small-bowel disease or resection with an intact colon → markedly increased oxalate absorption.**
- **Transport: paracellular (passive) and transcellular (largely active, carrier-mediated).**
 - **Solute-linked carrier family 26 (SLC26) anion exchangers** — **SLC26A1 and SLC26A6** mediate **intestinal oxalate secretion** (loss of function would therefore raise net absorption); shown in **rodents, not yet in humans**.
- **Modifiers of absorption:**
 - **Oxalate-binding cations (calcium, magnesium)** — **co-ingestion of calcium with oxalate forms insoluble calcium oxalate**, limiting free oxalate.
 - **Oxalate-degrading bacteria** — **Oxalobacter formigenes** uses oxalate as its **energy source**; in colonised rats it also **stimulates colonic secretion of endogenous oxalate**, lowering urinary oxalate by more than luminal degradation alone.
 - *O. formigenes* is the **most potent** degrader but **other species also degrade oxalate**; in healthy people it sits within a **cooperative bacterial network**. **Its net contribution to stone risk is not fully understood.**
- **Absorbed oxalate is almost entirely excreted in urine.**
- **Sources of urinary oxalate: about half dietary** (varying with calcium and oxalate intake), the rest **endogenous hepatic synthesis**.
- **Liver = primary site of synthesis**; the **primary hyperoxaluria (PH)** enzyme defects are hepatic and increase glyoxylate, the direct precursor of oxalate.
- **Endogenous sources:**
 - **Ascorbic acid** — the **largest endogenous source**, up to **40% of urinary oxalate**, via **non-enzymatic** conversion to **2,3-diketogulonic acid** → oxalate plus threosone or erythrulose.
 - **Hydroxyproline** (collagen breakdown; **gelatin**) — about **15% of endogenous synthesis**.
- **Renal handling: 86–98% of plasma oxalate is ultrafilterable; tubular secretion (probably proximal)** is supported by animal and clinical data; **SLC26A6** is implicated; the candidate **sulphate anion transporter-1 (SAT1, SLC26A1)** showed **no correlation** with hyperoxaluria in rats. **Whether stone formers handle oxalate differently is unknown.**

Concept — *Calcium and oxalate are handled in mirror image: calcium is well absorbed and tightly regulated, oxalate is poorly absorbed, unregulated and fully excreted — so urinary oxalate simply tracks what the gut lets through (free luminal oxalate, colonic exposure, bacterial degradation) plus hepatic synthesis.*

KEY POINTS: MINERAL METABOLISM

- **Calcium absorption is primarily small-intestinal** and depends on intake.
- **1,25-Dihydroxyvitamin D3 is the most potent stimulator** of intestinal calcium absorption.
- **PTH stimulates 1 α -hydroxylase** in the proximal tubule to convert 25-hydroxyvitamin D3 to 1,25(OH)₂D3.
- **PTH enhances renal calcium reabsorption and phosphate excretion.**

- **Intestinal oxalate absorption is modulated by luminal calcium, magnesium and oxalate-degrading bacteria.**

PATHOPHYSIOLOGY OF UPPER URINARY TRACT CALCULI

Classification of Nephrolithiasis

- **Drug-related stones** (triamterene, silica, indinavir, ephedrine) are **uncommon and usually preventable**.
- **Classification is by underlying metabolic or environmental abnormality** (Table 95.2); **calcium stones** arise from **hypercalciuria, hypocitraturia, hyperuricosuria and hyperoxaluria**, alone or combined.
- **Non-calcium stones form in unique settings:**
 - **Uric acid** — only in acid urine.
 - **Cystine** — impaired renal cystine reabsorption.
 - **Infection (struvite)** — alkaline urine from urease-producing bacteria.
- **Stone composition alone can direct therapy for some stones (e.g. cystine), but calcium stones need metabolic and environmental evaluation** because causes are multiple. **Molecular and genetic insights may reshape treatment.**

TABLE 95.1 STONE COMPOSITION AND RELATIVE OCCURRENCE

Group	Stone composition	Occurrence (%)
Calcium-containing	Calcium oxalate	60
	Hydroxyapatite (mixed with calcium oxalate in text)	20
	Brushite	2
Non-calcium	Uric acid	7
	Struvite	7
	Cystine	1–3
	Triamterene; silica; 2,8-dihydroxyadenine	< 1 each

From Pearle MS, Pak CYC, 1996 (*International Yearbook of Nephrology*).

- **Purpose:** the **frequency hierarchy** of stone types — the denominator for "commonest" questions.
- **Pattern:** **calcium stones** $\approx$ **80%** (oxalate $\gg$ apatite $\gg$ brushite); **uric acid and struvite tie at $\approx$ 7%**; **cystine is the rarest common metabolic stone**; drug and enzyme-defect stones $<$ **1%**.
- **Trap:** hydroxyapatite's 20% in the table refers to **mixed calcium oxalate–hydroxyapatite** stones in the text — **pure apatite is not the second commonest stone**.

TABLE 95.2 DIAGNOSTIC CLASSIFICATION OF NEPHROLITHIASIS

Condition	Metabolic / environmental defect	Prevalence (%)
Absorptive hypercalciuria	Increased gastrointestinal calcium absorption	20–40
— Renal phosphate leak	Impaired renal phosphate reabsorption	(within absorptive)
Renal hypercalciuria	Impaired renal calcium reabsorption	5–8
Resorptive hypercalciuria	Primary hyperparathyroidism	3–5
Hyperuricosuric calcium nephrolithiasis	Dietary purine excess, uric acid overproduction	10–40

Condition	Metabolic / environmental defect	Prevalence (%)
Hypocitraturic calcium nephrolithiasis	Isolated (idiopathic); chronic diarrhoea (gastrointestinal alkali loss); distal RTA (impaired renal acid excretion); thiazide-induced (hypokalaemia)	10–50
Hyperoxaluric calcium nephrolithiasis	Primary (oxalate overproduction); dietary (increased dietary oxalate); enteric (increased intestinal oxalate absorption)	2–15
Hypomagnesiuric calcium nephrolithiasis	Decreased intestinal magnesium absorption	5–10
Idiopathic low urine pH	Low urinary pH	15–30
Cystinuria	Impaired renal cystine reabsorption	< 1
Infection stones	Urease-producing bacteria	1–5
Low urine volume	Inadequate fluid intake	10–50
Miscellaneous or no abnormality	—	< 3

Modified from Pearle MS, Pak CYC, 1996 (*International Yearbook of Nephrology*).

- **Purpose:** the **metabolic work-up map** — each 24-hour urine abnormality points to a condition and a defect.
- **Pattern:** hypercalciuria (absorptive) is the single largest identifiable category; hypocitraturia and low volume have the widest ranges (10–50%); < 3% of formers have no abnormality — i.e. almost every stone former has a findable cause.
- **Trap:** thiazides cause hypocitraturia via hypokalaemia — the stone-preventing drug can itself lower citrate.
- **Trap:** primary hyperparathyroidism explains only 3–5% of calcium stone formers.

STONE FREQUENCIES — VALUES COMPARED ACROSS THE CHAPTER

Stone	Table 95.1	Table 95.2	Body text	Reason for difference
Uric acid	7%	— (idiopathic low pH 15–30%)	8–10% in US; up to 25% in parts of Germany; rising (7 → 14%)	Different series, eras and regions
Struvite / infection	7%	1–5%	5–15%	Different series and eras; incidence declining
Cystine	1–3%	< 1%	≈ 1% (adults); up to 10% of paediatric stones	Adult versus paediatric denominator
Calcium (all)	82%	—	≈ 80%	Rounding

Concept — Classification is by the urinary abnormality, not the stone — calcium stones in particular are a final common product of many different derangements, so the stone's composition starts the work-up rather than ending it.

Calcium Stones

Hypercalciuria

- **Hypercalciuria is the commonest abnormality in calcium stone formers (35–65%),** though its role is debated because **urinary calcium overlaps** between formers and non-formers.

- **Evidence for a causal role:**

- **Treatment that lowers urinary calcium reduces recurrence; persistent hypercalciuria predicts medical failure.**
- In three large cohorts with 24-hour urines, **incident stone risk rises with urinary calcium** after adjustment.
- **Randall plaque number correlates with urinary calcium** and with the number of stone episodes.
- **Mechanism:** high urinary calcium → **↑ saturation of calcium salts** and **↓ inhibitor activity** (calcium complexes negatively charged inhibitors such as **citrate and chondroitin sulphate**).
- **Normal load:** kidney filters ≈ **270 mmol calcium/day** and reabsorbs **all but ≈ 4 mmol**.

Definition of hypercalciuria	Threshold
Strictest	> 200 mg/day after 1 week of 400 mg calcium, 100 mg sodium diet
Weight-based / sex-based	> 4 mg/kg/day, or > 7 mmol/day (males), > 6 mmol/day (females)

- **Any threshold is artificial** — urinary calcium exerts a **continuous, graded effect** on risk.
- **Pak classification (1974)** — three subtypes by organ of dysregulation (**intestine, kidney, bone**): **absorptive, renal and resorptive**.
 - Critics: hypercalciuria reflects **multiple interrelated disturbances**; single **gene mutations affect several organs**; the classification **has not improved therapeutic outcomes** and is **not routinely used in practice**. It is retained in the chapter as the standard framework.

PAK CLASSIFICATION OF HYPERCALCIURIA — COMPARISON

Feature	Absorptive (AH)	Renal	Resorptive
Primary defect	↑ Intestinal calcium absorption	Renal calcium leak (↓ tubular reabsorption)	↑ Bone resorption — primary hyperparathyroidism
Share of stone formers	≈ 30% (Table 95.2: 20–40%)	5–8%	3–5% (≈ 5%)
Serum calcium	Normal	Normal	High (or high-normal)
Serum PTH	Normal or suppressed	↑ (secondary hyperparathyroidism)	↑ (primary)
Fasting urinary calcium	Normal (< 0.11 mg/dL glomerular filtrate)	High (≥ 0.11 mg/dL glomerular filtrate)	High
Urinary calcium after oral calcium load	≥ 0.2 mg calcium/mg creatinine	—	—
Serum phosphorus	Normal (low in renal phosphate leak)	Normal	Low
Sequence	Hyperabsorption → transient serum calcium rise → PTH suppressed → ↑ filtered calcium → hypercalciuria	Renal loss → PTH ↑ → ↑ calcitriol → gut absorption and bone resorption compensate	PTH ↑ → bone resorption + ↑ calcitriol → gut absorption → hypercalcaemia
Discriminating test	Normal fasting calcium + suppressed PTH	High fasting calcium + high PTH, normal serum calcium	High serum calcium + inappropriately high PTH; thiazide challenge

Absorptive Hypercalciuria

- **Definition and discriminators in the table;** severe forms may also show **fasting hypercalciuria**.

- **Type I versus type II:**
 - **Type II — dietary calcium restriction normalises urinary calcium.**
 - **Type I — does not normalise with restriction.**
- **Cause of hyperabsorption is unresolved:** vitamin D-dependent and -independent processes and **VDR up-regulation** proposed; **no mechanism explains all findings.**
- **Genetic candidates:**
 - **CYP24A1** (encodes **1,25(OH)2D-24-hydroxylase**, which inactivates 25(OH)D and 1,25(OH)2D3) — **biallelic loss-of-function → elevated calcitriol**, particularly with high vitamin D intake.
 - Causes **idiopathic infantile hypercalcaemia** (autosomal recessive; vitamin D supplement sensitivity).
 - In adults: **hypercalcaemia, hypercalciuria, nephrocalcinosis and nephrolithiasis.**
 - **Carrier frequency 4–20%, but not all affected individuals manifest disease.**
 - GWAS links CYP24A1 variants to **serum vitamin D levels.**
 - **CYP27B1** — generates calcitriol by 1-hydroxylation of 25(OH)D (context for the above).
 - **Vitamin D hypersensitivity** increases absorption; **VDR gene polymorphism** and **linkage to the VDR locus (chromosome 12q12–q14)** reported in familial hypercalciuria — but **not confirmed** by others.
 - **Chromosome 1q23.3–q24** locus for an inherited form of AH, with a putative gene **homologous to rat soluble adenylate cyclase.**
- **Renal phosphate leak** (a subtype of AH):
 - **Sequence:** renal phosphate wasting → hypophosphataemia → ↑ calcitriol → ↑ gut calcium and phosphate absorption → hypercalciuria.
 - **Hereditary hypophosphataemic rickets with hypercalciuria (HHRH)** — **autosomal recessive**; candidate genes **SLC34A1** and **SLC34A3** (encoding **NaPi-IIa** and **NaPi-IIc**, apical proximal tubule sodium–phosphate co-transporters).
 - **Rare — at most 2–4% of stone formers.**

Renal Hypercalciuria

- The kidney filters ≈ **270 mmol calcium** and must reabsorb > **98%**; ≈ **70% is reabsorbed proximally**, mostly paracellularly.
- **Fasting urinary calcium plus PTH separates a renal from an intestinal cause** (table).
- **Cause unknown**; monogenic disorders illuminate the transcellular and paracellular pathways:

MONOGENIC DISORDERS OF RENAL CALCIUM HANDLING

Disorder	Gene / protein	Site	Key features
Dent disease (X-linked recessive nephrolithiasis)	Chloride channel-5 (ClC-5)	Proximal tubule, TAL, α-intercalated cells	Hypercalciuria, low-molecular-weight proteinuria, nephrolithiasis, nephrocalcinosis, progressive renal failure
Familial hypomagnesaemia with hypercalciuria and nephrocalcinosis (FHHNC)	Claudin-16 (paracellin-1) and claudin-19	TAL, DCT	Loss of voltage-driven paracellular magnesium and calcium reabsorption
Claudin-14 variants	Claudin-14	TAL	Associated with nephrolithiasis

Disorder	Gene / protein	Site	Key features
Bartter syndrome (autosomal recessive group)	SLC12A1 (NKCC2, Na⁺-K⁺-2Cl⁻ co-transporter); KCNJ1 (ROMK apical potassium channel); CLCNKB (ClC-Kb basolateral chloride channel); BSND (barttin, subunit of ClC-Ka/ClC-Kb)	TAL	Salt wasting, hypokalaemic metabolic alkalosis , variable hypercalciuria and nephrolithiasis
Bartter type V	Potent activating CaSR mutation	TAL	Salt-losing nephropathy, secondary hyperaldosteronism (ROMK dysfunction from constitutive CaSR activation)
Autosomal dominant hypocalcaemia	Activating CaSR mutation (lower set-point)	Parathyroid, TAL	↓ PTH, ↓ renal calcium reabsorption → hypocalcaemia with hypercalciuria
CaSR loss-of-function SNPs (two synonymous SNPs reducing renal CaSR mRNA)	CaSR	TAL	Normocitraturic idiopathic nephrolithiasis : ↑ TAL paracellular reabsorption → interstitial calcium precipitation, hypocalciuria ; ↓ distal calcium delivery may impair acidification and concentration → calcium oxalate stones
NKCC2 and ROMK mutations (autosomal recessive)	NKCC2, ROMK	TAL	Fasting hypercalciuria and nephrocalcinosis

- **CaSR set-point mutations in CASR, GNA11 (G-protein α11) or AP2S1 (adaptor-related protein complex 2 σ1) shift calcium sensing → hyper- or hypocalcaemic disorders.**
- **CaSR controls TAL calcium reabsorption through paracellular permeability.**

Concept — Renal hypercalciuria is a leak with compensation: the kidney loses calcium, PTH rises to defend serum calcium — so the fasting urine is high and PTH is high while serum calcium is normal, the exact mirror of absorptive hypercalciuria (fasting urine normal, PTH suppressed).

Resorptive Hypercalciuria

- **Infrequent; most commonly primary hyperparathyroidism (PHPT), the cause of ≈ 5% of nephrolithiasis.**
- **Normocalcaemic variant: normal serum calcium with inappropriately high PTH** makes diagnosis harder.
- **Thiazide challenge:** thiazide ↑ renal calcium reabsorption → **exacerbates hypercalcaemia without suppressing PTH** in PHPT — unmasks the diagnosis.
- **Nephrolithiasis occurs in < 5% of patients with PHPT.**
- **Suspect PHPT when a stone former has serum calcium > 10.1 mg/dL.**
- **Serum calcium varies by up to 5% and mild PHPT raises it only slightly → repeat serum calcium with intact PTH; ionised calcium** helps in equivocal cases (may be high when total calcium is normal).
- **PTH also causes proximal bicarbonate and phosphate loss → phosphaturia and mild hyperchloraemic acidosis.**

Hypercalcaemia-Induced Hypercalciuria

- **Rare causes: malignancy, sarcoidosis, thyrotoxicosis, vitamin D toxicity.**

HYPERCALCAEMIC CAUSES OF HYPERCALCIURIA — COMPARISON

Cause	Mechanism	PTH	Distinguishing feature
Primary hyperparathyroidism	Autonomous PTH	High / inappropriately normal	Commonest cause of hypercalcaemia in outpatients; thiazide challenge
Sarcoidosis (and other granulomas)	Macrophage 1 α -hydroxylase → extrarenal calcitriol → gut absorption and bone resorption (absorptive + resorptive)	Suppressed	Rapid correction with corticosteroids; alveolar cells and lymph nodes synthesise calcitriol
Malignancy	PTH-related protein (PTHrP), osteolytic cytokines, direct bone destruction	Suppressed (intact PTH low)	Commonest cause of hypercalcaemia in hospitalised patients
Glucocorticoid excess	↑ Bone resorption, ↓ bone formation, ↑ PTH release; but ↓ gut calcium absorption	Stimulated	Stones common in Cushing syndrome

- **Granulomatous diseases** reported to cause hypercalcaemia: **tuberculosis, sarcoidosis, histoplasmosis, leprosy, silicosis** — **sarcoidosis is the one most associated with urolithiasis.**
- **Malignancy-associated hypercalcaemia:**
 - **Tumour mix:** lung and breast ≈ 60%; renal cell 10–15%; head and neck ≈ 10%; haematological (lymphoma, myeloma) ≈ 10%.
 - **Humoral factors:** PTHrP (production regulated by tumour-cell CaSR), **transforming growth factor, interleukin-1, tumour necrosis factor** → osteoclast activation.
 - **PTHrP mechanism:** like PTH, **↑ renal calcium reabsorption**; stimulates osteoblasts to secrete **receptor activator of nuclear factor- κ B ligand (RANKL)** → binds **RANK** on osteoclast precursors → mature osteoclasts → bone resorption; also **stimulates 1 α -hydroxylation** → calcitriol → gut absorption.
- **Glucocorticoids:**
 - **Most potent effect on bone** — **↑ resorption, ↓ formation** → osteopenia; **↑ PTH release.**
 - **↓ Intestinal calcium absorption** — the basis of steroid efficacy in sarcoid hypercalciuria.
 - **Net effect favours stones:** stones in **about half of active Cushing syndrome**, fewer when cured, fewest in controls; active disease shows **hypercalciuria, hypocitraturia and hyperuricosuria** — but **obesity and diabetes confound.**

Concept — The PTH level sorts the hypercalcaemias: high or inappropriate in primary hyperparathyroidism, suppressed in everything else (sarcoid calcitriol, tumour PTHrP) — and corticosteroids, which cut gut calcium absorption, are the specific remedy for the calcitriol-driven (granulomatous) form.

Hyperoxaluria

- **Definition:** urinary oxalate > 40 mg/day.
- **Effects:** **↑ calcium oxalate saturation**; also implicated in **crystal growth and retention via tubular cell injury** — **lipid peroxidation and oxygen free radicals** → membrane injury → **crystal fixation.**
 - **Antioxidants** prevent calcium oxalate precipitation in rat kidney and reduce oxalate excretion; **free-radical scavengers (phytic acid, mannitol)** prevent crystal deposition in vitro.
 - **But in humans**, large oxalate loads (up to 8 mmol) **did not raise oxidative-stress or renal-injury markers** — the importance of oxalate-induced membrane damage is questioned.
- **Three causes:** **primary (biosynthetic enzyme defect), enteric (malabsorption — inflammatory bowel disease, coeliac disease, intestinal resection), dietary (excess intake or substrate).**

Primary Hyperoxaluria

- **Rare autosomal recessive disorders of glyoxylate metabolism:** glyoxylate cannot be converted to **glycine**, so it is **oxidised to oxalate**, a metabolic end product.
- **Urinary oxalate markedly high (≥ 100 mg/day)** → calcium oxalate crystals in tubular lumens → some **attach to epithelium and aggregate into stones**, others are **internalised and extruded into the interstitium** → **marked nephrocalcinosis**.
- **Renal injury:** direct toxicity of oxalate or crystals (**reactive oxygen species**), **recurrent obstructing stones**, and **inflammation and interstitial fibrosis** from nephrocalcinosis.
- **Vicious circle:** falling renal function → oxalate elimination fails → **systemic oxalosis** (systemic calcium oxalate deposition).

PRIMARY HYPEROXALURIA TYPES 1–3 — COMPARISON

Feature	PH1	PH2	PH3
Enzyme	Alanine-glyoxylate aminotransferase (AGT) — pyridoxal phosphate (vitamin B6)-dependent	Glyoxylate reductase/hydroxypyruvate reductase (GRHPR)	4-Hydroxy-2-oxoglutarate aldolase (HOGA)
Gene	AGXT (≥ 253 mutations)	GRHPR (≈ 60 mutations)	HOGA1
Organelle	Liver peroxisome	Liver	Mitochondria
Urinary markers	Oxalate + glycolate	Oxalate + L-glyceric acid	Oxalate; severe hypercalciuria
Course	Most aggressive — ESRD in second to third decade; untreated → ESRD inevitable (88% by age 60 in the Rare Kidney Stone Consortium, RKSC)	Less aggressive renal course	Recurrent childhood stones that may become clinically silent ; only one reported case of ESRD
Specific therapy note	Pyridoxine-responsive in Gly170Arg (with Pro11Leu) mistargeting	—	Mechanism of hyperoxaluria not established
Transplant	Combined liver–kidney for most severe cases	Isolated kidney transplant, needed less often	Rarely reaches ESRD

- **PH1 mistargeting mutation:** commonest AGXT mutation **Gly170Arg**; with the **Pro11Leu minor-allele polymorphism**, AGT is **sent to mitochondria (inactive) instead of peroxisomes** → **pyridoxine responds** because its product **pyridoxal phosphate** is the AGT cofactor, raising catalytic activity and **restoring peroxisomal targeting**.
- **PH2 biochemistry:** ↓ GRHPR → ↑ hydroxypyruvate and glyoxylate → **lactate dehydrogenase (LDH)** converts them to **L-glyceric acid and oxalate**.
- **PH3 biochemistry:** HOGA converts hydroxy-2-oxoglutarate (from **hydroxyproline**) to pyruvate and glyoxylate.
- **Nephrocalcinosis on imaging** → **about 70% higher ESRD risk (RKSC)**.
- **Why liver–kidney transplant in PH1:** the **liver is the only organ that detoxifies glyoxylate** — a kidney alone would be re-injured by continuing hepatic oxalate overproduction.
- **Genetic testing** (all three types): confirms biochemistry, allows **prenatal diagnosis**, and identifies affected relatives.
 - **All three types can stay asymptomatic until adulthood** → **screen family members**.
 - **Exclude PH in relatives before living-related kidney donation**.
 - **Caveat: not all carriers develop PH**.

- **SLC26A1 (SAT1 sulphate-oxalate transporter) mutations** — another inherited hyperoxaluria; consider when PH1–3 are excluded.

Concept — All three primary hyperoxalurias block the disposal of glyoxylate in the liver, forcing it to oxalate; severity follows the enzyme (PH1 $\gg$ PH2 > PH3), and because the defect is hepatic, only replacing the liver cures PH1.

Enteric Hyperoxaluria

- The commonest cause of acquired hyperoxaluria.
- Mechanism (Fig. 95.9) — a sequence: fat malabsorption $\rightarrow$ luminal fatty acids saponify calcium and magnesium $\rightarrow$ less calcium available to complex oxalate $\rightarrow$ more free luminal oxalate $\rightarrow$ colonic hyperabsorption; malabsorbed fatty acids and bile salts also $\uparrow$ colonic permeability to oxalate.
- Fecal fat correlates strongly with urinary oxalate in steatorrhoea (Fig. 95.10).
- Co-existing risks in chronic diarrhoea: dehydration, hypokalaemia, hypomagnesiuria, hypocitraturia, low urine pH.
- Any malabsorption — small-bowel resection, intrinsic small-bowel disease, jejunioileal bypass — can cause it.
- Requires an intact colon — the colon is the principal site of oxalate hyperabsorption; in terminal-ileal Crohn disease stone incidence rises with length of bowel resected, provided the colon is intact.
- Cystic fibrosis: increased stone risk (overall prevalence about 4.5%); proposed causes — enteric hyperoxaluria from fat malabsorption, loss of O. formigenes, hyperuricosuria, hypocitraturia.
- Bariatric surgery:
 - Jejunioileal bypass was abandoned partly for renal failure and stones from severe hyperoxaluria.
 - Roux-en-Y gastric bypass (RYGB) — initially considered safer, but oxalate nephropathy with dialysis or transplantation was reported; retrospective, cross-sectional and prospective studies confirm $\uparrow$ urinary oxalate after RYGB and other malabsorptive procedures.
 - Hyperoxaluria develops ≥ 6 months after surgery.
 - Falling urinary calcium and uric acid partially offset the oxalate rise — net effect on calcium oxalate saturation varies.
 - Nonetheless stone incidence is higher after RYGB than in obese controls (claims data).
 - Cause: loss of O. formigenes is unlikely (colonisation is already low before surgery and does not differ from obese controls); an exaggerated urinary response to oral oxalate load points to malabsorption / enteric hyperoxaluria.

Bariatric procedure	Type	Stone risk
Long-limb Roux-en-Y gastric bypass; duodenal switch	Malabsorptive	Increased (enteric hyperoxaluria)
Gastric banding; sleeve gastrectomy	Restrictive	Not increased

Governing contrast	Colon intact (short bowel, small-bowel disease)	Colon absent (ileostomy after colectomy)
Oxalate	Hyperoxaluria (colonic hyperabsorption)	No hyperoxaluria
Typical stone	Calcium oxalate	Uric acid and ammonium acid urate (low volume, low pH, low sodium)

Dietary Hyperoxaluria

- Oxalate-rich foods: nuts, chocolate, brewed tea, spinach, potatoes, beets, rhubarb — overindulgence causes hyperoxaluria in otherwise normal people.

- **Dietary oxalate contributes 24–42% of urinary oxalate** (in this analysis).
- **Severe calcium restriction ↑ oxalate absorption** (less luminal binding) — **calcium restriction is counter-productive**.
- **Ascorbic acid supplements ↑ urinary oxalate by in vivo conversion**, but **clinical stone risk is not unequivocally increased**.

Role of the Microbiome

- **Humans lack oxalate-degrading enzymes**; some gut bacteria degrade oxalate.
- ***O. formigenes* uses oxalate as its sole carbon source** — a potential **modulator of intestinal oxalate**.
- **Stone formers have reduced or absent colonisation**, and **non-colonised individuals have higher urinary oxalate**.
- **Antibiotic exposure depletes *O. formigenes***:
 - **Cystic fibrosis** (prolonged antibiotics) — **absent *O. formigenes*, raised urinary oxalate**.
 - ***Helicobacter pylori* eradication** — **lower colonisation** than untreated *H. pylori* controls.
- **Population case-control study (UK electronic records)**: exposure to **any of five antibiotic classes 3–12 months earlier ↑ stone risk** — yet *O. formigenes* is **resistant to some of those classes**, so the effect may reflect **reduced overall microbiome diversity**, not *O. formigenes* alone.

Idiopathic Hyperoxaluria

- **Mild hyperoxaluria may matter as much as hypercalciuria** in idiopathic calcium oxalate stones.
- **Arabian Peninsula**: calcium stone prevalence **far higher than in the West** despite **near-absent hypercalciuria** — **hyperoxaluria is the predominant risk factor**.
- **Abnormal oxalate metabolism and transport** may contribute.

Concept — *Outside the rare primary forms, hyperoxaluria is a problem of free luminal oxalate reaching the colon — raised by fat malabsorption, low calcium intake, high oxalate intake or loss of degrading bacteria — so colonic presence and luminal calcium are the levers.*

Hyperuricosuria

- **Definition**: urinary uric acid > 600 mg/day.
- **Isolated in up to 10% of calcium stone formers**; combined with other abnormalities in up to 40%.

HOW URIC ACID PROMOTES CALCIUM OXALATE STONES (FIG. 95.11) — THREE MECHANISMS

Phase	Mechanism	Effect
I. Colloid	Colloidal sodium urate/uric acid adsorbs macromolecular inhibitors	Inhibitor removal → calcium oxalate nucleation
II. Solution	"Salting out" — dissolved urate salt lowers calcium oxalate solubility	↑ Calcium oxalate precipitation
III. Crystalline	Monosodium urate or uric acid crystals act as nidi	Heterogeneous nucleation / epitaxial growth of calcium oxalate

- Each mechanism has supporting evidence; which dominates is unclear.
- **Commonest cause**: dietary purine excess.
- **Other causes**: gout; myeloproliferative and lymphoproliferative disorders; multiple myeloma; secondary polycythaemia; pernicious anaemia; haemolytic disorders; haemoglobinopathies and thalassaemia; complete or partial hypoxanthine-guanine phosphoribosyltransferase (HGPRT) deficiency; phosphoribosylpyrophosphate synthetase overactivity; hereditary renal hypouricaemia.

- **URAT1** (urate-anion exchanger, **proximal tubule**, gene **SLC22A12**): **mutations → hyperuricosuric hypouricaemia (renal uric acid leak), exercise-induced acute renal failure and high stone risk.**
- **Evidence against a role:** in three large cohorts with 24-hour urines, **urinary uric acid was inversely associated with incident stones in males**, marginally inversely in younger females and not associated in older females.
- **Evidence for — landmark trial, Ettinger (1986):** randomised, placebo-controlled trial of **allopurinol** in **hyperuricosuric, normocalciuric calcium oxalate stone formers — more than twofold reduction in recurrence. But the benefit cannot be attributed with certainty to lowering urinary uric acid.**

Concept — *Uric acid drives calcium oxalate stones through urate's physical chemistry (inhibitor adsorption, salting out, epitaxy) rather than through uric acid stones themselves — and although epidemiology is conflicting, the allopurinol trial shows that in selected hyperuricosuric formers treatment works.*

Figure legends

- **Fig. 95.8 — Hepatic oxalate pathway:** glyoxylate → glycine by AGT (vitamin B6) — **blocked in PH1**; glyoxylate → glycolate / hydroxypyruvate → D-glycerate by GRHPR — **blocked in PH2**; hydroxyproline → 4-hydroxy-2-oxoglutarate → glyoxylate + pyruvate by HOGA — **defective in PH3**; LDH converts glyoxylate to oxalate and hydroxypyruvate to L-glycerate.
- **Fig. 95.10 — Fecal fat versus urinary oxalate in steatorrhoea:** linear relationship (oxalate $\approx 2.1 \times$ fecal fat + 30.7; r^2 0.4), on diets of **300–500 mg oxalate and 500–900 mg calcium** per day. (From Worcester, 1996.)

Hypocitraturia

- **An important and correctable abnormality:** isolated in up to **10%** of calcium stone formers; **combined with other abnormalities in 20–60%.**
- **Actions of citrate (four):**
 1. **Complexes calcium** → ↓ saturation of calcium salts.
 2. **Prevents spontaneous nucleation** of calcium oxalate.
 3. **Inhibits agglomeration and sedimentation** of calcium oxalate crystals, and **growth** of calcium oxalate and calcium phosphate crystals.
 4. **Enhances the inhibitory effect of Tamm-Horsfall glycoprotein** at normal levels.
- **Definition:** urinary citrate < 320 mg/day.
- **Acid-base status is the primary determinant of citrate excretion:** **metabolic acidosis → ↑ tubular citrate reabsorption and ↓ peritubular citrate synthesis → hypocitraturia.**
- **The defect is renal, not intestinal:**
 - Stone formers and controls have **similar serum citrate and filtered load**, but formers show **lower urinary citrate and higher tubular reabsorption.**
 - **Oral citrate is rapidly and almost completely absorbed** (96–98% within 3 hours) in both hypocitraturic formers and controls — so **absent overt bowel disease, gut malabsorption is not the cause.**

Cause of hypocitraturia	Mechanism
Distal RTA	Systemic acidosis — urine pH high (> 6.8 as quoted here), high serum chloride, low serum bicarbonate and potassium; confirmed by failure to acidify after ammonium chloride load
Chronic diarrhoea	Intestinal alkali loss → systemic acidosis
Excess animal protein	Acid load; high-protein, low-carbohydrate diets ↓ citrate and pH (low citrus, high animal protein)
Thiazides	Hypokalaemia → intracellular acidosis

Cause of hypocitraturia	Mechanism
Angiotensin-converting enzyme (ACE) inhibitors	Hypocitraturia independent of systemic acidosis or hypokalaemia — perhaps intracellular acidosis
Strenuous exercise	Lactic acidosis
Isolated (idiopathic)	Unrelated to acidosis

- Citrate rises with alkalosis and with PTH, oestrogen, magnesium, calcitonin and vitamin D.

Concept — Citrate excretion is a readout of intracellular pH in the proximal tubule: anything that acidifies the cell (systemic acidosis, hypokalaemia, acid diet) makes the tubule reclaim citrate — which is why alkali (potassium citrate) is the treatment.

Low Urine pH

- Below pH 5.5, undissociated uric acid predominates → uric acid and/or calcium stones; calcium oxalate forms by heterogeneous nucleation on uric acid crystals.
- Any disorder that lowers urine pH predisposes to stones.
- Chronic metabolic acidosis → low urine pH, hypercalciuria (↑ bone resorption and renal calcium leak) and hypocitraturia.
- Idiopathic low urine pH (formerly "gouty diathesis") — stone propensity with low urine pH of unknown cause, with or without gouty arthritis.

Renal Tubular Acidosis

- RTA = metabolic acidosis from defective tubular hydrogen-ion secretion or bicarbonate reabsorption. Three types: 1, 2 and 4.
- Type 1 (distal) matters most to urologists: commonest form and most stone-associated (stones in up to 70%); stone symptoms lead to the diagnosis in up to half of cases.

Normal renal acidification

- Bicarbonate is freely filtered; the kidney must reclaim ≈ 4500 mmol/day, mainly proximally, and excrete the daily acid load (from carbohydrate, fat and protein metabolism and stool bicarbonate loss) distally.

Governing contrast	Proximal tubule	Distal tubule / collecting duct
Job	Reclaim filtered bicarbonate (the bulk)	Net acid excretion (plus 5–10% of bicarbonate)
Machinery	Carbonic anhydrase generates $\text{H}^+ + \text{HCO}_3^-$; apical Na^+/H^+ exchanger (driven by basolateral Na^+, K^+ -ATPase); basolateral $\text{Na}^+/\text{HCO}_3^-$ co-transporter; apical H^+ -ATPase; luminal carbonic anhydrase converts H_2CO_3 to $\text{H}_2\text{O} + \text{CO}_2$	α -Intercalated cells: apical H^+ -ATPase and H^+, K^+ -ATPase; basolateral $\text{Cl}^-/\text{HCO}_3^-$ exchanger (AE1)
Net effect	Bicarbonate absorbed without net H^+ secretion or pH change	H^+ excreted bound to buffers — titratable acid (mainly phosphate) and ammonia (as NH_4^+); 1000:1 gradient → urine pH as low as 4.5
Luminal carbonic anhydrase	Present	Absent — prevents rapid carbonic acid dissociation, helping maintain the gradient

- Both proximal and distal RTA produce hyperchloraemic metabolic acidosis with inappropriately high urine pH; the distinction is acid excretion (distal, type 1) versus bicarbonate reabsorption (proximal, type 2).

Type 1 (Distal) Renal Tubular Acidosis

- **Collecting-duct failure to acidify despite systemic acidosis: hypokalaemic, hyperchloraemic, non-anion-gap metabolic acidosis, nephrolithiasis, nephrocalcinosis, urine pH > 6.0.**
- **Incomplete distal RTA: cannot lower urine pH below 5.5 after an acid load, but no systemic acidosis — serum electrolytes normal.**
- **Stone type: calcium phosphate, from hypercalciuria, hypocitraturia and high urine pH.**
 - **Acidosis → bone demineralisation → secondary hyperparathyroidism → hypercalciuria.**
 - **Profound hypocitraturia — probably the most important factor —** from acidosis, abnormal tubular citrate transport, or citrate movement into mitochondria with intracellular acidosis.
- **Forms: hereditary, idiopathic or acquired; most sporadic; autosomal dominant and recessive patterns exist.**
 - **Inherited forms: growth retardation, nephrocalcinosis, calculi, hypokalaemic metabolic acidosis.**
 - **Autosomal recessive form: more severe, earlier onset, with learning disability and sensorineural hearing loss.**
 - **Children are one-third of cases —** vomiting, diarrhoea, failure to thrive, growth retardation.
 - **Secondary distal RTA: Sjögren syndrome, systemic lupus erythematosus, obstructive uropathy, pyelonephritis, acute tubular necrosis, hyperparathyroidism, idiopathic hypercalciuria.**
- **Molecular basis — α -intercalated cell dysfunction (Fig. 95.12):**

Gene	Protein	Inheritance	Features
SLC4A1	AE1 $\text{Cl}^-/\text{HCO}_3^-$ exchanger	Mostly autosomal dominant; complete or incomplete	Hearing loss not usually a feature
ATP6V1B1	B1 subunit of H^+ -ATPase	Autosomal recessive	Heterozygous B1 carriers may have a hypofunctional pump → incomplete distal RTA
ATP6V0A4	A4 subunit of H^+ -ATPase	Autosomal recessive	—
CA2	Carbonic anhydrase II (proximal tubule, loop, α -intercalated cells)	Recessive	Mixed proximal–distal RTA: bicarbonate wasting, cannot acidify below pH 5.5, ↓ NH_4^+ excretion

Type 2 (Proximal) Renal Tubular Acidosis

- **Defect in proximal bicarbonate reabsorption — a self-limiting sequence: proximal HCO_3^- loss → distal tubule overwhelmed → bicarbonaturia, high urine pH → plasma HCO_3^- falls → filtered HCO_3^- falls → distal capacity suffices → steady state.**
- **Usually part of a Fanconi-type generalised proximal defect —** loss of glucose, protein (amino acids), uric acid and phosphate.
- **Clinical: growth retardation and hypokalaemia in children.**

Type 4 Renal Tubular Acidosis

- **Chronic renal damage —** interstitial disease, diabetic nephropathy.
- **Low urine pH, decreased acid excretion; ↓ GFR → hyperkalaemic, hyperchloraemic metabolic acidosis (urinary HCO_3^- loss, ↓ ammonium excretion).**
- **Aldosterone resistance is common → ↓ distal acidification and H^+/K^+ exchange → ↓ ammoniogenesis, worse hyperkalaemia.**
- **Can still acidify urine after an acid challenge.**
- **Stones uncommon —** reduced calcium and uric acid excretion from impaired renal function.

- But low urine pH (from hyperkalaemia-impaired ammoniogenesis) predisposes to uric acid stones.

RENAL TUBULAR ACIDOSIS TYPES 1, 2 AND 4 — COMPARISON

Feature	Type 1 (distal)	Type 2 (proximal)	Type 4 (hyperkalaemic)
Defect	Distal H ⁺ secretion (α-intercalated cell)	Proximal HCO ₃ ⁻ reabsorption	Aldosterone deficiency/resistance, ↓ ammoniogenesis
Serum potassium	Low	Low	High
Urine pH	Always high (> 6.0); cannot go < 5.5	High initially, then < 5.5 at steady state	Low; can acidify
Serum HCO ₃ ⁻	Low (may be very low)	15–18 mEq/L	Low
Urinary citrate	Profoundly low	Relatively normal	—
Stones	Common (up to 70%) — calcium phosphate; nephrocalcinosis	Uncommon	Uncommon — but uric acid if any
Associations	Sjögren, lupus, obstruction, hypercalciuria; genetic (SLC4A1, ATP6V1B1, ATP6V0A4)	Fanconi syndrome	Diabetic nephropathy, interstitial disease

Concept — Stone risk in RTA follows citrate and urine pH: only distal (type 1) RTA combines fixed alkaline urine with profound hypocitraturia and hypercalciuria, the recipe for calcium phosphate stones; proximal RTA spares citrate, and type 4 lowers the stone load (except uric acid, via low pH).

Hypomagnesiuria

- **Rare:** isolated in < 1% of stone formers; combined with other abnormalities in 6–11%.
- **Mechanism:** magnesium complexes oxalate and calcium salts, so low magnesium means less inhibition; low urinary magnesium also tracks low urinary citrate (cause or effect unclear).
- **Causes:** poor dietary intake, intestinal malabsorption from chronic diarrhoeal syndrome.
- **Evidence is contradictory:**
 - Rat studies both implicate and question magnesium.
 - Hypomagnesiuric patients had higher recurrence in one study; others found no difference in magnesium excretion between formers and controls — possibly because few formers are hypomagnesiuric.
- **Magnesium raises urinary pH, citrate and magnesium and lowers calcium oxalate saturation** in vitro and in vivo, but two randomised trials of magnesium oxide versus placebo or no treatment showed no clinical benefit.

Concept — Magnesium is a physicochemically plausible but clinically unproven inhibitor: favourable urine chemistry has not translated into fewer stones in randomised trials.

Uric Acid Stones

- **Humans and apes lack hepatic uricase** (which converts uric acid to allantoin) — the gene was silenced (pseudogenised) in evolution, leaving higher blood and urine uric acid.
- **Allantoin is 10–100 times more soluble than uric acid** — hence human susceptibility to uric acid stones.
- **Frequency:** 8–10% of US stones, up to 25% in parts of Germany; rising — a US registry saw the proportion double (≈ 7% → 14%) over 30 years; a Norwegian series rose from 2% to 9% over 40 years.
- **Uric acid is a weak acid, pKa 5.35 at 37 °C** — at that pH, half is urate salt, half free acid.

- **Sodium urate is about 20 times more soluble than free uric acid → the free-acid fraction (set by pH) determines risk.**
- **pH governs solubility** (Fig. 95.13):
 - At pH 5, even modest uric acid exceeds solubility.
 - At pH 6.5, > 1200 mg/L remains soluble.
 - **Solubility limit of undissociated uric acid ≈ 96–100 mg/L** — readily exceeded by normal daily excretion (500–600 mg/day), so urine is often supersaturated **below pH 6**.
- **Low pH also drives calcium oxalate and mixed stones** — uric acid and sodium urate act as **nidi** (heterogeneous nucleation, epitaxy).
- **Uric acid stone formation after precipitation is poorly understood**; crystal adhesion and glycosaminoglycans are proposed but unproven — **no natural urinary inhibitor of uric acid is known**.
- **Theobromine** (dimethylxanthine in **cocoa and chocolate**) **inhibits uric acid nucleation and growth in vitro**; cocoa products **raised urinary theobromine and reduced uric acid crystallisation** in volunteers (in vitro urine assay).

THE THREE DETERMINANTS OF URIC ACID STONE FORMATION (FIG. 95.14)

Determinant	Causes	Importance
Low urine pH	High animal protein diet, chronic diarrhoea, primary gout, obesity ↔ insulin resistance	The single most important factor — most uric acid formers have normal uric acid excretion but persistently low pH
Low urine volume	Diarrhoeal states, volume depletion, hot climate or occupation	Concentrates uric acid
Hyperuricosuria	High animal protein diet, primary gout, myeloproliferative disorders, uricosuric drugs, congenital disorders	Least often the driver in idiopathic cases

- **Drosophila model**: knockout of an orthologue of human **Na⁺/H⁺ exchanger regulatory factor (NHERF1)** → more acidic urine and more uric acid crystals in **Malpighian tubules** — underlining pH.
- **Aetiological categories**:
 - **Congenital** — defects in **renal urate transport or uric acid metabolism** → hyperuricosuria.
 - **Acquired** — **chronic diarrhoea, volume depletion, myeloproliferative disorders, high animal protein intake, uricosuric drugs** — may act on any of the three determinants.
 - **Idiopathic** — **decreased fractional excretion of urate, no gout**.

Governing contrast	Idiopathic uric acid nephrolithiasis	Hyperuricosuric calcium nephrolithiasis
Urinary uric acid	Normal	High
Urine pH	Low (acidic)	Normal
Other urine features	Low ammonium, high titratable acid	High urinary sodium and calcium → ↑ sodium urate and calcium oxalate saturation
Stone	Uric acid	Calcium oxalate

Low Urine pH

- **Pathogenesis uncertain and probably multifactorial**; the leading model links it to **insulin resistance**:
 - **Normouricosuric pure uric acid formers are more often diabetic or glucose-intolerant**.

- On a **controlled metabolic diet** their **urine pH is lower** than controls or calcium oxalate formers; they excrete **less acid as ammonium, more as titratable acid and less citrate** — **impaired ammonium excretion**.
- **Uric acid stones are far commoner in type 2 diabetes** (about a third of stones in diabetics versus roughly a tenth or less in non-diabetics in two series).
- Uric acid formers share **metabolic syndrome features** — **hypertriglyceridaemia, hyperglycaemia, obesity, hypertension**.
- **Hyperinsulinaemic-euglycaemic clamp**: low urine pH correlates with insulin resistance in normal subjects, and **uric acid formers are the most insulin-resistant**.
- **Hyperglycaemia** → lower urine pH and higher uric acid saturation.
- **Body weight is inversely related to urine pH, independent of urinary sulphate** (animal-protein marker); **BMI ≥ 27 kg/m² predicts urine pH ≤ 5.5** .
- **Mechanism (Fig. 95.15)**: insulin promotes proximal tubular ammoniagenesis from glutamine and stimulates the Na^+/H^+ exchanger NHE3, which carries NH_4^+ or supplies luminal H^+ to trap NH_3 → insulin resistance → ↓ ammonium production/excretion → unbuffered H^+ → low urine pH.
- **Diet-independent impairment**: higher BMI → lower ammonium-to-sulphate ratio (ammoniogenesis corrected for acid intake) and **higher uric acid supersaturation**.
- **Increased endogenous acid production**:
 - On a fixed **low acid-ash diet**, uric acid formers have **higher net acid excretion**.
 - **Controlled for urinary sulphate**, net acid excretion is higher in uric acid formers and in **non-forming type 2 diabetics**.
 - Matched for age, sex and BMI on a metabolic diet, uric acid formers show **lower pH, higher titratable acid and net acid excretion**, and **higher uric acid supersaturation despite equal uric acid excretion**.
- **Synthesis**: with **impaired ammonium excretion plus increased endogenous acid**, **titratable acids become the main urinary buffer**; acid-base balance is kept, **but at a lower urine pH** than ammonium (higher pKa) would allow.
- **Why ammoniogenesis fails — renal lipotoxicity**:
 - **Metabolic switch**: proximal tubule uses **free fatty acids instead of glutamine** → less ammoniogenesis.
 - **Lipotoxicity**: fat deposited in non-adipose tissue injures cells; in the **Zucker diabetic fatty (ZDF) rat**, **renal steatosis reduced NHE3 expression and activity**.
 - Uric acid stone matrix proteomics: proteins mainly of **inflammation and complement**; pathways chiefly **phospholipid and fatty acid metabolism**.
 - **Visceral fat on CT** is associated with **low urine pH (< 6)** and independently with **uric acid composition**.
- **Diet**: in a randomised crossover of **vegetable, vegetable-plus-egg and animal protein diets** (rising sulphate), **animal protein raised urinary calcium**, gave the **most undissociated uric acid and the least citrate** (lower pH); matched for electrolytes and protein, it **raised uric acid stone risk but — because of opposing factors — not calcium oxalate or calcium phosphate risk**.

Concept — *Uric acid stones are a disease of urine pH, not of uric acid: insulin resistance and renal fat impair ammoniogenesis, so the kidney buffers acid with titratable acid at a lower pH — making the uric acid stone a renal manifestation of metabolic syndrome.*

Hyperuricosuria

- **Definition**: **urinary uric acid > 600 mg/day** — causes supersaturation with undissociated uric acid.
- **Gout**: stones **markedly fewer** with urinary uric acid **< 600 mg/day** than with **> 1000 mg/day**.

- **Causes:** dietary, acquired and hereditary diseases, **urate-transporter defects** (as above).

Low Urinary Volume

- **Any cause of low volume raises uric acid supersaturation** — **hot-temperature workers** show high uric acid supersaturation; **warm climates (e.g. Israel)** have **high uric acid stone rates**.

Cystine Stones

- **Normal handling:** amino acids are freely filtered and **almost completely reabsorbed proximally**.
- **Transporter:** cystine and the other **dibasic amino acids** cross the apical membrane of the **proximal tubule and jejunum** via a **sodium-independent heteromeric amino acid transporter**, exchanging for neutral amino acids.

Subunit	Gene	Chromosome	ICC type
rBAT (heavy)	SLC3A1 ($\approx$ 247 mutations)	2p (short arm of chromosome 2)	Type A
b ⁰ , ⁺ AT (light)	SLC7A9 ($\approx$ 160 mutations)	19q (long arm of chromosome 19)	Type B
Both	—	2 and 19	Type AB

- **Mutation in either subunit → cystinuria.**
- **Inheritance:** autosomal recessive (rarely autosomal dominant with incomplete penetrance).
- **No nutritional consequence** of the identical jejunal defect — amino acids are absorbed as **oligopeptides and whole proteins**.
- **Fractional excretion of cystine:** $\approx$ 0.4% normally, up to 100% in cystinuria.
- **Only cystine forms stones** — other dibasic amino acids are soluble; **cystine = two cysteines joined by a disulphide bond**, far less soluble than cysteine; intracellular reduction to cysteine maintains the reabsorption gradient.
- **Epidemiology:** incidence 1 in 1000 to 1 in 17,000 (US and Europe); up to 10% of paediatric stones; more frequent and more severe in males.
- **Solubility determinants:** cystine concentration, pH, ionic strength, urinary macromolecules.
 - **Supersaturation is the main driver** — there is no specific urinary inhibitor of cystine crystallisation.
 - **Precipitates at physiological urine conditions.**
 - **Highly pH-dependent:** $\approx$ 300 mg/L at pH 5, $\approx$ 400 mg/L at pH 7, $\approx$ 1000 mg/L at pH 9.
 - **Ionic strength** (0.005 → 0.3) dissolves up to $\approx$ 70 mg/L more; **colloids** also increase solubility → **cystine is more soluble in urine than in synthetic solution** (Fig. 95.16).
- **Co-existing abnormalities** in cystine formers — **hypocitraturia (commonest)**, **hyperuricosuria**, **hypercalciuria** — can produce **calcium or mixed calcium-cystine stones**.
- **Classification:**
 - **Historical** — by urinary cystine in **obligate heterozygote parents**; **correlated poorly with genotype**.
 - **International Cystinuria Consortium (ICC)** — by chromosome: **type A (chromosome 2)**, **type B (chromosome 19)**, **type AB (both)**.
- **ICC data:** homozygotes excrete up to $\approx$ 2000 μ mol cystine/g creatinine; mean age at first stone $\approx$ 12 years; stone episodes $\approx$ 0.42/year in males versus 0.21/year in females.
- **Heterozygotes:** **type B carriers excrete far more cystine** ($\approx$ 475 versus $\approx$ 70 μ mol/g creatinine in type A), yet stone formation does not differ and is uncommon in type B carriers.

- **Renal function:** stone formers generally have **higher CKD risk**, and **cystine formers have lower creatinine clearance than other formers**; in a French cohort about a quarter of adults had **eGFR < 60 mL/min/1.73 m²** and many had **hypertension**.
 - Explanation: **more open surgery, including nephrectomy**; histology shows **dilated ducts of Bellini plugged with cystine, glomerulosclerosis and interstitial fibrosis**.

Concept — *Cystinuria is a pure supersaturation disease — a single transport defect, no inhibitor, and a solubility that only rises meaningfully at alkaline pH — so only dilution and a markedly alkaline pH move it out of the precipitating range, and repeated surgery erodes renal function.*

Infection Stones

- **Composition:** **magnesium ammonium phosphate hexahydrate (struvite, $\text{MgNH}_4\text{PO}_4 \cdot 6\text{H}_2\text{O}$)**, often with **carbonate apatite ($\text{Ca}_{10}[\text{PO}_4]_6 \cdot \text{CO}_3$)**.
- **History (sequence of discovery):** a Swedish geologist found magnesium ammonium phosphate in **guano** and named it **struvite** after his mentor **H.C.G. von Struve** → bacteria proposed to **split urea** (early 20th century), with **Proteus vulgaris** isolated from a stone → a **bacterial enzyme** proposed to hydrolyse urea → **urease isolated** (from **Canavalia ensiformis**, the jack bean) → now established that **struvite forms only with urea-splitting infection**.

Pathophysiology

- **Urease is absent from sterile human urine — infection with a urease-producing organism is a prerequisite.**
- **Chemical cascade (step → step):**
 1. **Urea + H_2O → 2NH_3 + CO_2** (bacterial urease).
 2. **Alkaline urine (pH 7.2–8.0) → NH_3 + H_2O → NH_4^+ + OH^-** (pKa 9.0). Normally alkalinity would halt ammonium generation, but **urease keeps producing ammonia**, pushing pH higher.
 3. **CO_2 hydration → H_2CO_3 → H^+ + HCO_3^-** (pKa 6.3) → **H^+ + CO_3^{2-}** (pKa 10.2).
 4. **Phosphate dissociation: H_2PO_4^- → H^+ + HPO_4^{2-}** (pKa 7.2) → **H^+ + PO_4^{3-}** (pKa 12.4).
 5. With physiological **magnesium** → **struvite**; with **calcium, phosphate and carbonate** → **carbonate apatite and hydroxyapatite** (Fig. 95.17).
- **Obstruction or stasis** may associate with or exacerbate infection stones → **growth can be rapid**.

Bacteriology

- **Enterobacteriaceae** make up most urease producers, but many **gram-positive and gram-negative bacteria, some yeasts and Mycoplasma species** make urease (Table 95.3).
- **Commonest urease producers: Proteus, Klebsiella, Pseudomonas and Staphylococcus; Proteus mirabilis is the organism most associated with infection stones.**
- **Escherichia coli** — a common uropathogen, but **only rare strains produce urease**.
- **Yet contemporary series culture E. coli (and sterile urine) from struvite patients surprisingly often; E. coli matched Proteus as the commonest isolate in one international series; higher struvite content predicts urease-producing organisms.**
- **Explanation: bladder urine and stone cultures are frequently discordant → voided urine culture is unreliable for identifying the stone organism.**

TABLE 95.3 ORGANISMS THAT MAY PRODUCE UREASE

Group	Usually urease-positive (> 90% of isolates)	Occasionally urease-positive (5–30% of isolates)
Gram-negative	Proteus (rettgeri, vulgaris, mirabilis, morganii); Providencia stuartii ; Haemophilus influenzae; Bordetella pertussis; Bacteroides corrodens; Yersinia enterocolitica; Brucella species; Flavobacterium species	Klebsiella pneumoniae , Klebsiella oxytoca ; Serratia marcescens ; Haemophilus parainfluenzae; Bordetella bronchiseptica; Aeromonas hydrophila; Pseudomonas aeruginosa ; Pasteurella species
Gram-positive	Staphylococcus aureus ; Micrococcus; Corynebacterium (ulcerans, renale, ovis, hofmannii)	Staphylococcus epidermidis ; Bacillus species; Corynebacterium (murium, equi); Peptococcus asaccharolyticus; Clostridium tetani; Mycobacterium rhodochrous group
Mycoplasma	T-strain Mycoplasma ; Ureaplasma urealyticum	—
Yeasts	Cryptococcus; Rhodotorula; Sporobolomyces; Candida humicola; Trichosporon cutaneum	—

From Gleeson MJ, Griffith DP, 1990 (Urolithiasis: a medical and surgical reference).

- **Purpose:** which organisms can make urease — i.e. which infections can build a struvite stone.
- **Pattern:** **Proteus** is reliably urease-positive; **Klebsiella** and **Pseudomonas** only occasionally; **Staphylococcus aureus** usually, **S. epidermidis** occasionally; **Ureaplasma** is a reliable, easily missed producer.
- **Trap:** **E. coli** does not appear — yet it is frequently cultured from struvite patients (discordant cultures, or rare urease-positive strains).

Pathogenesis

- **Beyond chemistry,** urease-producing bacteria **damage the urothelial mucosal layer** → ↑ bacterial colonisation and **crystal adherence**.
- **Ammonium** from ureolysis **alters the protective glycosaminoglycan layer** → ↑ bacterial adherence → **biofilm**.
- **Bacteria live within an exopolysaccharide matrix; struvite crystallises inside and around bacterial cells** → stone growth, **bacteria encased within the stone**.

Epidemiology

- **5–15% of all stones.**
- **Staghorn calculi are not mostly infection stones:** in a modern analysis **less than half were infection stones**; the majority were **metabolic**, most often **calcium phosphate**.
- **Incidence is falling worldwide** (Australia, Japan; in the US **in females**, little change in males) — attributed to **better diagnosis and treatment of urinary tract infection (UTI) in females** and **more effective stone surgery**.
- **Geography:** highest in **India and Pakistan** (poor sanitation → female UTI); **lowest in Canada, China, Argentina, Poland, Iraq and Italy**.
- **Female-to-male ratio ≈ 2:1.**
- **At-risk groups:** elderly; premature infants and congenital urinary tract malformation; diabetes; urinary stasis from obstruction, urinary diversion or neurological disorders.

- **Spinal cord injury** — risk of **infection and metabolic stones** (neurogenic dysfunction plus **immobilisation hypercalciuria**); **functionally complete cord transection** → **highest staghorn risk**.

Concept — *Infection stones need one thing no human urine can supply — urease; everything else (alkaline pH, ammonium, carbonate, phosphate) follows chemically, and biofilm then locks the bacteria inside the stone.*

Miscellaneous Stones

Xanthine and Dihydroxyadenine Stones

- **Autosomal recessive purine disorders**: enzyme deficiency causes accumulation of the precursor **immediately proximal** to the block.

Contrast	Xanthine	2,8-Dihydroxyadenine (DHA)
Enzyme deficient	Xanthine dehydrogenase (XDH)	Adenine phosphoribosyltransferase (APRT)
Accumulating compound	Xanthine	Adenine → converted by XDH to DHA
Gene	—	APRT, chromosome 16q24; > 40 mutations (Met136Thr common in Japanese; an intron-4 splice-site T insertion common in a French cohort)
Radiology	Radiolucent — mistaken for uric acid	Radiolucent — mistaken for uric acid
Solubility	Highly insoluble	Extremely insoluble at any pH
Iatrogenic cause	High-dose allopurinol (XDH inhibition) — e.g. Lesch-Nyhan syndrome (HGPRT deficiency)	—
Treatment clue	—	Allopurinol prevents; febuxostat more effective at studied doses

- **DHA** → stones, obstructive nephropathy or crystalline nephritis.
- **Allopurinol rarely causes xanthine stones** — only **partial XDH inhibition**; serum uric acid rarely falls below 3 mg/dL.
- **APRT deficiency can present in infancy** and may be confused with **HGPRT deficiency** because **DHA** resembles uric acid.
- **Rare Kidney Stone Consortium APRT-deficiency registry**: allopurinol 400 mg/day and febuxostat 80 mg/day both lowered DHA excretion; febuxostat more effective.

Ammonium Acid Urate Stones

- ≈ 1% of stones; endemic bladder stones in children in developing countries.
- Radiolucent.
- **Associations**: chronic diarrhoea, inflammatory bowel disease, ileostomy, laxative abuse, recurrent UTI, recurrent uric acid stones, morbid obesity; a contemporary series added **prostate surgery with bladder neck contracture** and found **no laxative association**.
- **Ileostomy after colectomy**: low urine volume, pH and sodium, and **no hyperoxaluria** (no colon) → **ammonium acid urate** and **uric acid stones**, not calcium oxalate; inflammatory bowel disease with ileostomy gave the **highest ammonium urate content**.
- **Laxative abuse mechanism** (sequence): **gastrointestinal fluid loss** → **dehydration** → **intracellular acidosis** → ↑ **urinary ammonia**; low urinary sodium leaves urate to **complex with ammonium** → ammonium acid urate supersaturation.

- **Precipitation favoured at pH 6.2–6.3.**
- **Link with uric acid stones:** shared **low volume and low pH**; mixed uric acid–ammonium urate stones are usually **uric acid-predominant**; fluctuations in acidity, ammonium and sodium shift the balance between uric acid and sodium- or ammonium-bound urate.
- **Obesity (BMI ≥ 30)** was the **commonest feature** once bowel-disease patients were excluded; **BMI correlated with ammonium urate content** — consistent with the obesity–low urine pH link.

Matrix Stones

- **Protein suspensions can promote calcium stones; osteopontin and calprotectin** help build the matrix of calcium stones.
- **Predominantly matrix stones are rare, radiolucent, and may be mistaken for tumour or uric acid stones;** literature is **mostly case reports**.
- **Matrix is $\approx 2.5\%$ of calcium stone dry weight; pure matrix stones contain $> 65\%$ protein.**
- **Composition $\approx$ two-thirds mucoprotein, one-third mucopolysaccharide;** matrix of crystalline stones closely resembles that of matrix stones.
- **Why they fail to calcify is unclear** — low urinary calcium was proposed, but **patients had normal urinary calcium**.
- **Dialysis patients:** proteinuria may contribute; matrix contains **microfibrillar protein and β_2 -microglobulin**.
- **Proteus infection:** electron microscopy shows **fibrous net-like laminations with bacterial, cellular and crystalline material**; proteomics found proteins **mostly previously unreported**, the majority **inflammatory or defensive**.

Medication-Related Stones

- **Two routes:** **direct** (the drug or a metabolite **precipitates**) or **indirect** (the drug **alters urine chemistry** to favour metabolic stones).
- **Stones of the drug itself** (with excessive intake): **ephedrine, triamterene, guaifenesin, silicate, indinavir, ciprofloxacin**.
- **Calcium-stone promoters:** **loop diuretics** (furosemide, bumetanide) and **carbonic anhydrase inhibitors** (acetazolamide, topiramate, zonisamide).

Medications That Directly Promote Stone Formation

Antiretroviral Agents

- **Indinavir sulphate** — protease inhibitor that **raises CD4⁺ counts** and **lowers HIV RNA** in human immunodeficiency virus (HIV) infection and **acquired immunodeficiency syndrome**.
 - **Stone incidence 4–13%.**
 - **Pharmacokinetics:** peak plasma level **< 1 hour**; hepatic metabolism, mainly faecal elimination, but **about half excreted unchanged in urine**.
 - **pKa 5.5; solubility falls steeply with rising pH:** ≈ 0.300 mg/mL at pH 5, 0.035 at pH 6, 0.020 at pH 7 — **far more soluble below pH 5.5**.
 - **Standard dose $\rightarrow$ urinary concentration near solubility limit ≈ 3 hours after ingestion** at average volume and pH.
 - **Crystalluria in about two-thirds** of new users; stable at about a quarter of sediments after 2 weeks.
 - **Now infrequently used.**

ANTIRETROVIRAL AGENTS AND STONE FORMATION — COMPARISON

Agent	Class	Urinary excretion unchanged	Stone risk	Note
Indinavir	Protease inhibitor	≈ 50%	4–13% (highest)	Least soluble at physiological pH; acidification increases solubility ; now rarely used
Ritonavir-boosted atazanavir	Protease inhibitor	≈ 7%	≈ 7% — highest of the newer agents	Widely used
Lopinavir-ritonavir	Protease inhibitor	—	Reported	—
Nelfinavir	Protease inhibitor	< 3%	Lower	—
Amprenavir	Protease inhibitor	< 3%	Lower	—

- **Rule: stone rate tracks urinary excretion of unchanged drug and its insolubility.**

Triamterene

- **Potassium-sparing diuretic** for hypertension; ≈ **0.4% of stones** in a large series, only a third largely triamterene.
- **Stone formers do not handle the drug differently** from controls; about half of all subjects exceed the solubility of the **sulphate metabolite**.
- **Tends to incorporate into existing stones or nidi rather than initiate stones** → rare in non-recurrent formers; **hospitalisation rates for stones were no different from hydrochlorothiazide users**.

Guaifenesin and Ephedrine

- **Large quantities** → **stones of their metabolites**; typically **over-the-counter cold remedies abused for ephedrine's stimulant effect**; drug-abuse history common.
- **Herbal ecstasy and ma huang** — ephedrine-containing preparations.
- **Tachyphylaxis** → **escalating doses**; toxicity includes **death, cardiomyopathy, stroke, hypertension, seizures**.

Silicate Stones

- **Silica** is ubiquitous (vegetables, grains, seafood, water) and easily excreted.
- **Silicate stones are extremely rare** — linked to **large intakes of silicate antacids (magnesium trisilicate)**.

Medications That Indirectly Promote Stone Formation**DRUGS THAT ALTER URINE CHEMISTRY — COMPARISON**

Drug / class	Urinary effect	Stone type
Corticosteroids, vitamin D, phosphate-binding antacids	Hypercalciuria	Calcium
Thiazides	Intracellular acidosis → hypocitraturia	Calcium
Loop diuretics (furosemide, bumetanide)	Inhibit TAL sodium and calcium reabsorption → hypercalciuria	Calcium oxalate — stones in up to 64% of low-birth-weight infants on furosemide
Carbonic anhydrase inhibitors	Metabolic acidosis with alkaline urine → hypocitraturia, hypercalciuria	Calcium phosphate
Laxatives (abuse)	Diarrhoea, low volume, low sodium, ammonia excretion, low pH	Ammonium acid urate

Drug / class	Urinary effect	Stone type
Cytotoxic agents	High cell turnover → uric acid excretion	Uric acid

CARBONIC ANHYDRASE INHIBITORS — COMPARISON

Agent	Indication	Carbonic anhydrase inhibition	Stone evidence
Acetazolamide	Classic carbonic anhydrase inhibitor	Strong ; blocks sodium bicarbonate reabsorption at multiple nephron sites	Calcium phosphate stones with chronic use
Topiramate	Epilepsy, migraine prophylaxis; off-label obesity, neuropathic pain, alcoholism, type 2 diabetes, smoking and cocaine dependence	Several isoenzymes	Label 1.2–1.5% is an underestimate : about one in ten users reported stones and one in five had asymptomatic stones on CT (mean ≈ 43 months); metabolic acidosis , high urine pH , low citrate , ↑ calcium phosphate saturation ; hypocitraturia dose-dependent and inversely related to duration
Zonisamide	Epilepsy (sulphonamide)	Weak	Increased stone risk

Concept — Drug stones follow two rules: for direct stones, risk equals urinary excretion of an insoluble drug (indinavir, atazanavir); for indirect stones, the drug simply recreates a known metabolic phenotype — carbonic anhydrase inhibitors mimic distal RTA (alkaline urine, low citrate, calcium phosphate), loop diuretics mimic hypercalciuria, laxatives mimic ileostomy.

Anatomic Predisposition to Stones

- **Anomalies causing obstruction or stasis carry high stone rates.** The long-running question: **stasis** (delayed transit → crystal formation and retention) or the same metabolic abnormalities as other stone formers?

Ureteropelvic Junction Obstruction

- **Stones in nearly 20% of patients with ureteropelvic junction obstruction (UPJO).**
- **Evidence that metabolic risk persists after the obstruction is corrected** (adult series):
 - **Most recurred after UPJO repair, and a large share of recurrences were in the contralateral kidney.**
 - **Most patients with non-infectious stones had a metabolic abnormality — a rate comparable to other stone formers.**
 - **The spectrum mirrored the general stone population — hypercalciuria commonest, then hypocitraturia, hyperuricosuria, primary hyperparathyroidism, RTA.**
 - **Treating the metabolic abnormality substantially cut recurrence** compared with conservative management.
- **Pre-repair metabolic evaluation:** UPJO patients with stones had **more metabolic abnormalities** (higher urinary calcium, **hypercalciuria and hyperuricosuria**) than UPJO patients without stones.
- **Children:**
 - **Stone-forming children — with or without UPJO — differ in urinary citrate and oxalate from non-stone-forming children with UPJO** (prospective comparison), supporting a metabolic contribution.
 - **UPJO confers a markedly increased (about seventyfold) stone risk** versus normal children.

- **Most children with non-struvite stones recurred after surgery**, and most had a metabolic abnormality; those who did **not** recur mostly **lacked** one.

Horseshoe Kidneys

- **Pooled stone incidence (meta-analysis):** $\approx 36\%$ in adults, $\approx 3\%$ in children.
- **High ureteral insertion** $\rightarrow$ **impaired drainage** $\rightarrow$ **UPJO tendency** — risk traditionally attributed to **stasis**.
- **But metabolic evaluation found at least one abnormality in every patient tested**; the distribution resembled normal-anatomy formers **except hypocitraturia, which was over-represented**.
- **Composition (meta-analysis):** calcium-based $\approx 89\%$, struvite $\approx 4\%$, uric acid $\approx 4\%$, other $\approx 3\%$.

Calyceal Diverticula

- **Stones in up to 40%.**
- **Conflicting evidence:**
 - One series: **metabolic abnormalities in half**, and **most had stones elsewhere** (synchronous or metachronous) — supports **metabolic risk**.
 - Another: **low metabolic abnormality rates** in patients with and without diverticula — but **pH and citrate were not measured**, so abnormalities were **under-counted** (low pH and hypocitraturia affect about 10% and 28% of recurrent formers).
 - **24-hour urine** of diverticular stone patients **resembled calcium oxalate formers** (hypercalciuria, $\uparrow$ calcium oxalate supersaturation) — yet **urine aspirated from the diverticulum had lower calcium oxalate supersaturation** than either renal pelvis.
- **Composition:** mainly calcium oxalate monohydrate; struvite-carbonate apatite when infected; concomitant UTI in up to 40% (*E. coli*, *Proteus*, *Pseudomonas*).

Medullary Sponge Kidney

- Medullary sponge kidney (MSK) = ectasia of the renal collecting ducts; nephrocalcinosis and calculi are frequent.
- **Candidate mechanisms:** infection and stasis in ectatic ducts, plus tubular defects — hypercalciuria, impaired concentrating ability, defective acidification after ammonium chloride load.
- **Acidification defects:** common on acid loading (mostly distal RTA, occasionally proximal); **impaired concentrating ability** in most; H^+ -ATPase gene mutations (ATP6V1B1, ATP6V0A4) found in MSK patients — supporting an MSK-distal RTA link.
- **Yet three studies of MSK patients with stones found no RTA.**
- **Hypercalciuria** is the commonest abnormality in some series (mostly **absorptive**), less frequent in others; sometimes **renal leak** from damaged tubules.
- **Hypocitraturia** the commonest abnormality in other series.
- **Urine proteomics** (Italian network) proposed biomarkers distinguishing MSK from idiopathic calcium formers — laminin $\alpha 2$, ficolin 1, mannan-binding lectin serine protease 2, complement component 4-binding protein, sphingomyelin, ephrins — role not yet established.

ANATOMIC ANOMALIES AND STONES — COMPARISON

Anomaly	Stone frequency	Metabolic abnormality	Distinctive finding	Verdict
UPJO	$\approx 20\%$ (children $\sim 70\times$ normal risk)	Common; same spectrum as general formers	Recurrence after repair, often contralateral	Metabolic > anatomic

Anomaly	Stone frequency	Metabolic abnormality	Distinctive finding	Verdict
Horseshoe kidney	≈ 36% adults, ≈ 3% children	Present in all tested	Hypocitraturia over-represented; calcium-based stones	Stasis + obligatory metabolic defect
Calyceal diverticulum	Up to 40%	Half in one series; under-measured in another	Diverticular urine less supersaturated than pelvic urine; UTI up to 40%	Metabolic + stasis
Medullary sponge kidney	Frequent (with nephrocalcinosis)	Hypercalciuria, hypocitraturia; distal RTA on acid loading	Ductal ectasia; H ⁺ -ATPase mutations	Metabolic even without RTA

Concept — Across every anomaly the answer is the same: anatomy sets the stage but metabolism writes the stone — which is why patients with anatomical stones still need a full metabolic evaluation, and why surgical correction alone does not prevent recurrence.

Stones in Pregnancy

- **Symptomatic stones:** 1 in 204 to 1 in 3000 pregnancies; incidence has not risen (1991–2000 versus 2001–2011), unlike the general population.
- **Timing:** mostly second and third trimesters, with flank pain or haematuria; odds rise in the second and third trimesters and peak 0–3 months post partum, returning to baseline by 1 year.
- **Misdiagnosis in up to 28%** — appendicitis, diverticulitis, placental abruption.
- **Physiological hydronephrosis in up to 90%**, persisting 4–6 weeks post partum:
 - Causes: progesterone and — at least contributory, perhaps primary — compression by the gravid uterus.
 - Right side greater — engorged uterine vein and dextrorotation of the uterus.
 - Dilatation may promote crystallisation through stasis; ↑ pelvic pressure may mobilise stones and provoke symptoms.

Pregnancy — opposing urinary effects		Mechanism
Promoters ↑		Renal blood flow ↑ → GFR ↑ 30–50% → ↑ filtered calcium, sodium, uric acid; placental 1,25(OH)2D3 → ↑ gut calcium absorption, PTH suppressed → absorptive-type hypercalciuria (reversible post partum); hyperuricosuria
Inhibitors ↑		↑ Citrate, magnesium and glycoproteins
Net		Overall stone risk similar to non-pregnant females

- **Composition:** some studies show no difference, but others find calcium phosphate predominant (about three-quarters in one multi-institutional series; significantly more carbapatite than in age-matched non-pregnant females) — perhaps from pregnancy-related urinary pH changes (speculative).

Concept — Pregnancy raises promoters and inhibitors together, so stone risk is roughly unchanged — but when stones do form they skew towards calcium phosphate, and the symptomatic peak comes late in pregnancy and just after delivery, when hydronephrosis and stasis are greatest.

KEY POINTS: PATHOGENESIS

- **Absorptive hypercalciuria:** normal serum calcium, normal or suppressed PTH, normal fasting urinary calcium, elevated urinary calcium.

- **Renal hypercalciuria:** impaired renal calcium reabsorption **stimulates PTH** and causes **fasting hypercalciuria**.
- **Resorptive hypercalciuria:** primarily **primary hyperparathyroidism**; also **granulomatous diseases** producing **1,25(OH)2D3**.
- **Low urinary pH is the most important determinant of uric acid stones.**
- **Low urine pH in uric acid formers** probably reflects **impaired ammoniagenesis from insulin resistance plus excess acid production**.
- **In distal RTA, a defective H⁺-ATPase prevents excretion of excess acid into the distal tubule.**
- **Infection stones require alkaline urine achievable only with urease-producing bacteria.**